



PROJECT COMPLETION REPORT



DESIGN ,SUPPLY, INSTALLATION AND COMMISSIONING OF WIFI NETWORK AND REVAMPING OF EXISTING LAN AT CEB HEAD OFFICE

Ceylon Electricity Board - Head Office

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Contents

1	Document Control.....	4
1.1	PROJECT COMPLETION REPORT	4
1.2	AUTHORIZATION	4
1.3	AMENDMENT RECORD.....	4
1.4	DISTRIBUTION LIST	4
2	Introduction	4
3	LAN Distribution	6
3.1	Huawei - AirEngine 5761-21 Access Point.....	8
3.2	Huawei - CloudEngine S5732-H48S6Q L3 Switch	9
3.3	Huawei CloudEngine S5735-L48LP4S-A-V2 Switch (PoE & Non-PoE Variants)	10
3.4	Huawei CloudEngine S5735-L24P4S-A-V2 Switch (PoE & Non-PoE Variants).....	11
3.5	Huawei iMaster NCE-Campus	12
4	Delivered Equipment List	13
5	Equipment Installed Location List with Serial Number	17
6	IP Address Plan.....	24
6.1	Devices Management IP Addresses	24
6.2	Network VLAN and Subnet Mapping	26
7	Network Design Diagram	28
8	Switch Port Zone – Core Switch Cluster	30
8.1	Core Switch	30
9	Configuration Information	31
9.1	Core Switch Configuration	32
9.2	Access Switch Configuration	48
9.3	iMaster Configuration (Sample)	61
10	Drawings	62
10.1.1	Head Quarters Building –Groud Floor.....	63
10.1.2	Head Quarters Building –First Floor.....	64
10.1.3	Head Quarters Building –Second Floor	65
10.1.4	Head Quarters Building –Third Floor	66
10.1.5	Head Quarters Building –Fourth Floor	67
10.1.6	Head Quarters Building –Fifth Floor	68
10.1.7	Head Quarters Building –Sith Floor.....	69
10.1.8	Head Quarters Building –Seventh Floor	70
10.1.9	Canteen Building –Ground Floor.....	71
10.1.10	Canteen Building –Second Floor	72

10.1.11	Canteen Building –Third Floor	73
10.2	Live Heat Maps.....	74
10.2.1	Heat Map - Head Quarters Building – Ground Floor 2.4Gz Band	75
10.2.2	Heat Map - Head Quarters Building – Ground Floor 5 Gz Band	76
10.2.3	Heat Map – Head Quarters Building – First Floor – 2.4 GHz Band	77
10.2.4	Heat Map – Head Quarters Building – First Floor – 5 GHz Band	78
10.2.5	Heat Map – Head Quarters Building – Second Floor – 2.4 GHz Band	79
10.2.6	Heat Map – Head Quarters Building – Second Floor –5 GHz Band	80
10.2.7	Heat Map – Head Quarters Building – Third Floor – 2.4 GHz Band.....	81
10.2.8	Heat Map – Head Quarters Building – Third Floor – 5 GHz Band.....	82
10.2.9	Heat Map – Head Quarters Building – Fourth Floor – 2.4 GHz Band	83
10.2.10	Heat Map – Head Quarters Building – Fourth Floor – 5 GHz Band	84
10.2.11	Heat Map – Head Quarters Building – Fifth Floor – 2.4 GHz Band.....	85
10.2.12	Heat Map – Head Quarters Building – Fifth Floor – 5 GHz Band.....	86
10.2.13	Heat Map – Head Quarters Building – Sixth Floor – 2.4 GHz Band	87
10.2.14	Heat Map – Head Quarters Building – Sixth Floor – 5 GHz Band	88
10.2.15	Heat Map – Head Quarters Building – Seventh Floor – 2.4 GHz Band	89
10.2.16	Heat Map – Head Quarters Building – Seventh Floor – 5 GHz Band	90
11	Administration Information	91
12	Warranty Certifications.....	92
12.2	Warranty Information	93
12.3	Contact Details	93
12.4	Support Contact Information	94
12.5	Support Escalation Information	94
12.6	Project Team	95
13	Certificate of Acceptance.....	96
14	User Guides and Administrative Guides	98
13	Notes	99
14	Amendments.....	100

1 Document Control

1.1 PROJECT COMPLETION REPORT

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1.2 AUTHORIZATION

	Prepared by	Reviewed by	Authorized by
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Designation	Associate Network System Engineer	Senior Network Systems Engineer	Consultant

1.3 AMENDMENT RECORD

Version	Issue date	Prepared by	Authorized by	Description
1.0	23.07.2025	Gayathra Hashan	Krishal Kahagalage	

1.4 DISTRIBUTION LIST

Name/Title	Organization	Comments
PCR	VSIS	
PCR	Ceylon Electricity Board - Head Office	

2 Introduction

This Project Completion Report presents a comprehensive overview of the CEB Head Office Wi-Fi Revamp Project, encompassing the design, supply, installation, commissioning, and maintenance of a state-of-the-art wireless and wired network infrastructure at the Ceylon Electricity Board Headquarters.

The deployment spans seven floors of the HQ building (Ground Floor through 7th Floor) and three floors of the canteen building (Ground, 2nd, and 3rd Floors), ensuring robust and seamless wireless coverage throughout critical operational areas. The network was meticulously planned and validated using a detailed site survey conducted with the Ekahau Sidekick, enabling optimal placement of access points and confirming comprehensive signal propagation tailored to the unique architectural features of the premises.

Centralized network management was achieved through a unified platform that integrates both wired and wireless components. The system leverages advanced features such as automated device provisioning, real-time performance monitoring, and security enforcement mechanisms including 802.1X-based Network Access Control, tightly integrated with RADIUS and Active Directory for robust user authentication and role-based access management.

The main server room, strategically located on the ground floor of the HQ building, serves as the nerve center for network operations, housing core switching infrastructure and management servers that support high availability and centralized control across all floors and buildings.

Comprehensive testing and validation procedures were undertaken post-installation, ensuring the network's stability, performance, and security met stringent project requirements. The project team also delivered targeted training sessions to both IT staff and end users, equipping them with the knowledge required to effectively operate and maintain the new network environment.

Extensive documentation, including system configuration files, operational procedures, and warranty certificates, has been handed over to the CEB IT department, facilitating ongoing support and future scalability.

This project delivers a scalable, secure, and high-performance enterprise network that aligns with CEB's digital transformation goals, providing a resilient foundation to support evolving business demands and enhanced service delivery.

3 LAN Distribution

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3.1 Huawei - AirEngine 5761-21 Access Point



The Huawei AirEngine 5761-21 access point leverages Wi-Fi 6 (802.11ax) technology combined with intelligent smart-antenna design to deliver high-performance connectivity in enterprise, educational, and retail environments. It operates concurrently on 2.4 GHz (2x2 MIMO) and 5 GHz (4x4 MIMO), achieving aggregate device throughput of up to 5.375 Gbps, significantly enhancing network efficiency in dense deployments. The integrated smart-antenna system dynamically steers beams toward active clients, improving coverage, reducing interference, and strengthening connectivity reliability. As an advanced Wi-Fi 6 AP, it supports MU-MIMO, OFDMA, 1024-QAM, BSS Coloring, and Target Wake Time to boost spectral efficiency, cut latency, and prolong mobile device battery life. The platform ensures robust security with WPA3/WPA2, 802.1X authentication, rogue-device detection, and dual Gigabit Ethernet ports. It also offers flexible deployment via PoE+ or DC power and supports Cloud/Fit/Fat management modes. USB and Bluetooth capabilities provide IoT expansion, smart O&M, and Bluetooth-based asset localization.

3.2 Huawei - CloudEngine S5732-H48S6Q L3 Switch



The Huawei CloudEngine S5732-H48S6Q is deployed as the core switch in this project, forming the backbone of the network infrastructure. It provides 44 GE SFP downlink ports, 4 × 10GE SFP+ uplinks, and 6 × 40GE QSFP+ uplinks, delivering 648 Gbps switching capacity and 486 Mpps forwarding performance. The switch incorporates an embedded wireless controller that supports up to 1,024 access points, enabling unified wired and wireless policy management and efficient CAPWAP processing. It also supports advanced Layer-2 and Layer-3 features including VXLAN with BGP-EVPN, iStack virtualization, IPv4/IPv6 routing (OSPF, BGP), MPLS, QoS, and the STP family, making it ideal for campus core deployment. For proactive monitoring and maintenance, the switch integrates real-time telemetry via iPCA and TWAMP, feeding into Huawei CampusInsight for early fault detection. Security is reinforced with support for 802.1X, DHCP snooping, ARP protection, MACsec, encrypted traffic analytics, and threat deception capabilities. Built on a 1 U rack-mount chassis with dual hot-swappable 600 W AC or 1 kW DC power modules and fans, it operates reliably between –5 °C and +45 °C—ensuring resilient performance in demanding enterprise environments.

3.3 Huawei CloudEngine S5735-L48LP4S-A-V2 Switch (PoE & Non-PoE Variants)



The Huawei CloudEngine S5735-L48LP4S-A-V2 switches were deployed as Layer 3 access switches with a mix of PoE and non-PoE models to support diverse endpoint requirements. The PoE variant provides 48 Gigabit Ethernet ports capable of delivering up to 380 W PoE power, suitable for powering devices such as wireless APs, IP phones, and surveillance cameras. The non-PoE variant offers the same port density but without power delivery, ideal for endpoints that do not require PoE. Both models support 4 GE SFP uplink ports, 104 Gbps switching capacity, and 78 Mpps forwarding rate. They offer comprehensive Layer 2/3 features including VLANs, static routing, OSPF, RIP, STP/RSTP/MSTP, QoS, DHCP snooping, and ACL-based security policies. The switches are housed in a 1U rack-mount chassis with integrated AC power supplies and dual fan cooling, operating reliably between -5°C and $+50^{\circ}\text{C}$. This flexible deployment approach ensures optimized network coverage and power delivery according to device needs.

3.4 Huawei CloudEngine S5735-L24P4S-A-V2 Switch (PoE & Non-PoE Variants)



The Huawei CloudEngine S5735-L24P4S-A-V2 switches were used as Layer 3 access switches with both PoE and non-PoE versions deployed based on endpoint power requirements. The PoE switches offer 24 Gigabit Ethernet PoE+ ports with a total PoE budget of approximately 370 W, suitable for powering wireless APs, IP cameras, and other devices. The non-PoE variants provide the same port density without power delivery for endpoints that do not require PoE. Both support 4 GE SFP uplinks, with 56 Gbps switching capacity and 42 Mpps forwarding performance. Full Layer 2 and Layer 3 functionalities are supported, including VLANs, static routing, OSPF, RIP, QoS, IGMP snooping, STP, and ACLs. They feature 1U rack-mount designs with integrated power supplies and fan cooling, operating within -5°C to $+50^{\circ}\text{C}$. This mixed deployment ensures efficient network segmentation and power provisioning tailored to device needs.

3.5 Huawei iMaster NCE-Campus

Huawei iMaster NCE-Campus (Version V300R024C00SPC101) is an AI-driven network management and control platform used in this project to centrally manage the entire campus network infrastructure. It provides a single-pane-of-glass interface to monitor, configure, secure, and optimize both wired and wireless networks. In this deployment, it was used to manage core and access switches, as well as wireless access points, simplifying day-to-day network operations and providing deep network visibility.

A major use case in the project was Network Access Control (NAC). The platform was configured to enforce 802.1X authentication across both wired and wireless clients. This authentication was done via RADIUS integration, using credentials from the organization's Active Directory (AD). This ensured that only authorized users and devices could connect to the network, based on predefined access policies. Different VLANs and access levels were assigned dynamically based on user identity or device type, improving both security and operational flexibility.

iMaster NCE-Campus also enabled real-time network monitoring, including device status, user behavior, and traffic analysis. The system uses AI and machine learning to detect anomalies, such as abnormal logins or rogue device activity, helping administrators identify potential threats early. It also offered graphical network topology views, wireless heatmaps, and location-based user tracking, making troubleshooting and performance optimization faster and more efficient.

In addition, iMaster NCE-Campus played a key role in automated configuration and policy deployment, including bulk device onboarding, firmware updates, and centralized push of VLAN and QoS settings. This reduced manual configuration errors and improved deployment speed. Its role-based access control allowed different levels of admin privileges for network operators and support teams, improving control over operations.

By integrating network management, security enforcement, and automation into a single platform, Huawei iMaster NCE-Campus provided a secure, scalable, and intelligent environment that meets modern enterprise network requirements. This implementation significantly reduced administrative overhead, improved network security, and delivered a seamless experience to end users..

4 Delivered Equipment List

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Huawei AirEngine 5761-21_Access Points					
No	Part Number	Model	Description	Unit Qty.	Qty.
1.1	WLAN AP Hardware				
	02354VQK	AirEngine5761-21	AirEngine5761-21(11ax indoor,2+4 dual bands,smart antenna,USB,BLE)	1	52
1.2	Power				
	02312WTB	HW-120200B1W	AC-DC 24W/12V Adaptor(UK)	1	52
1.3	Installation Accessories				
	21244035	B-3DT-INAP01	bracket accessory	1	52
	21155588	AP-2115B-7MM01	Thin mounting bracket	1	52

Huawei - CloudEngine S5735-L24P4S-A-V2 Switch					
No	Part Number	Model	Description	Unit Qty.	Qty.
2.1	S5735-L Series Mainframe				
	98012021	S5735-L24P4S-A-V2	S5735-L24P4S-A-V2 (24*10/100/1000BASE-T ports, 4*GE SFP ports, PoE+, AC power)	1	19
2.2	Optical Transceiver				
	GE-SFP Optical Transceiver				
	02313URD	eSFP-GE-SX-MM850	Optical Transceiver,eSFP,GE,Multi-mode Module(850nm,0.55km,LC)	1	19
3.3	Installation Materials				
	Patch Cord				
	14130222	SS-OP-D-LC-M-10	Patch cord-LC/PC-LC/PC-Multimode-A1b-2mm-10m-PVC-Orange	2	38

Huawei - CloudEngine S5732-H48S6Q Switch					
No	Part Number	Model	Description	Unit Qty.	Qty.
3.1	S5732 H Series Mainframe				
	02353AJU-004	S5732-H48S6Q	S5732-H48S6Q (44*GE SFP ports,4*10GE SFP+ ports,6*40GE QSFP ports,without power module)	1	2
3.2	Power				
	02312FFU-002	PAC600S12-EB	600W AC Power Module(Back to Front, Power panel side exhaust)	1	2
3.3	Optical Transceiver				
	GE-SFP Optical Transceiver				
	02313URG	SFP-1000BaseT	Electrical Transceiver,SFP,GE,Electrical Interface Module(100m,RJ45)	34	68
3.4	Installation Materials				
	21240537	MOUTEARB01	1U Boxlike equipment lengthening back mounting ear,IEC Expandable size(280-450mm)	1	2
	02232WFJ	EETSI1U01	Installation Kit for Fixture Box Platform 1U Chassis Installed in ETSI Rack	1	2
3.5	Stack Cables				
		QSFP_40G_C1M	SN: CN2049QN103L01V9 SN: CN2049QN103L01VA	1	2

Huawei – CloudEngine CloudEngine S5735-L24T4S-A-V2 Access Non-POE Switch					
No	Part Number	Model	Description	Unit Qty.	Qty.
4.1	S5735-L Series Mainframe				
	98012004	S5735-L24T4S-A-V2	S5735-L24T4S-A-V2 (24*10/100/1000BASE-T ports, 4*GE SFP ports, AC power)	1	1
4.2	Optical Transceiver				
	GE-SFP Optical Transceiver				
	02313URD	eSFP-GE-SX-MM850	Optical Transceiver,eSFP,GE,Multi-mode Module(850nm,0.55km,LC)	1	1
4.3	Installation Materials				
	Patch Cord				
	14130222	SS-OP-D-LC-M-10	Patch cord-LC/PC-LC/PC-Multimode-A1b-2mm-10m-PVC-Orange	2	2

Huawei-CloudEngine S5735-L48LP4S-A-V2 Access POE Switch					
No	Part Number	Model	Description	Unit Qty.	Qty.
5.1	S5735-L Series Mainframe				
	98012047	S5735-L48LP4S-A-V2	S5735-L48LP4S-A-V2 (48*10/100/1000BASE-T ports, 4*GE SFP ports, PoE+, AC power)	1	14
5.2	Optical Transceiver				
	GE-SFP Optical Transceiver				
	02313URD	eSFP-GE-SX-MM850	Optical Transceiver,eSFP,GE,Multi-mode Module(850nm,0.55km,LC)	1	14
5.3	Installation Materials				
	Patch Cord				
	14130222	SS-OP-D-LC-M-10	Patch cord-LC/PC-LC/PC-Multimode-A1b-2mm-10m-PVC-Orange	2	28

Huawei - CloudEngine S5735-L48T4S-A-V2 Access Non POE Switch					
No	Part Number	Model	Description	Unit Qty.	Qty.
6.1	S5735-L Series Mainframe				
	98012033	S5735-L48T4S-A-V2	S5735-L48T4S-A-V2 (48*10/100/1000BASE-T ports, 4*GE SFP ports, AC power)	1	3
6.2	Optical Transceiver				
	GE-SFP Optical Transceiver				
	02313URD	eSFP-GE-SX-MM850	Optical Transceiver,eSFP,GE,Multi-mode Module(850nm,0.55km,LC)	1	3
6.3	Installation Materials				
	Patch Cord	Patch Cord			
	14130222	SS-OP-D-LC-M-10	Patch cord-LC/PC-LC/PC-Multimode-A1b-2mm-10m-PVC-Orange	2	6

WAC 650					
No	Part Number	Model	Description	Unit Qty.	Qty.
7.1	AC6508 Hardware				
	02354FRJ-001	AC6508	AC6508 mainframe (10*GE ports, 2*10GE SFP+ ports, with the AC/DC adapter)	1	2
7.2	Delivery Accessories				
	21240477	E5700MK00	250mm*180mm*1Uequipment front mounting ear(1set)	1	2

iMaster NCE-Campus					
No	Part Number	Model	Description	Unit Qty.	Qty.
8.1	Customer-provided servers				
	05201140	OS-SUSE12SP5 Linux for PC	Operating System Software,SLES 12SP5,English Doc,1 Yr 7*24 Services,for PC Server,Only for Huawei Branded Products bundled with Huawei Software,Physical server,Multi-language,English Document,Paper License	3	3

5 Equipment Installed Location List with Serial Number

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Product	Version	Software ID	License SN
Huawei iMaster NEC-Campus	V300R024C00SPC101	2412-04BN-LP58	LIC20241204FVPF8T

Building Floor	Device Host Name	Device Model	SN
Main Building Ground Floor	Core_1 & 2	Huawei CloudEngine S5732 H48S6Q L3 Switch	102496854806 / 102496854816
	GF_SS_R1_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017647
	GF_SS_R2_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001332
	GF_RS_R3_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6000809
	GF_RS_R4_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017710
	HQ/GF/R01/W/01	Huawei AirEngine5761-21	2102354VQK10QA106091
	HQ/GF/R02/W/02	Huawei AirEngine5761-21	2102354VQK10QA106132
	HQ/GF/R03/W/03	Huawei AirEngine5761-21	2102354VQK10QA105953
	HQ/GF/R03/W/04	Huawei AirEngine5761-21	2102354VQK10QA105954
	HQ/GF/R04/W/05	Huawei AirEngine5761-21	2102354VQK10QA106264
	HQ/GF/R04/W/06	Huawei AirEngine5761-21	2102354VQK10QA105948
	HQ/GF/R03/W/07	Huawei AirEngine5761-21	2102354VQK10QA105950

Main Building First Floor	1F_RS_R5_SW1	Huawei CloudEngine L24T4S-A-V2 L3 Switch Non PoE	QU23C6048618
	1F_RS_R6_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6000719
	1F_RS_R7_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001378
	1F_SS_R8_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6000952
	1F_SS_R9_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001375
	HQ/1F/R07/W/08	Huawei AirEngine5761-21	2102354VQK10QA106250
	HQ/1F/R06/W/09	Huawei AirEngine5761-21	2102354VQK10QA106125
	HQ/1F/R07/W/10	Huawei AirEngine5761-21	2102354VQK10QA106093
	HQ/1F/R07/W/11	Huawei AirEngine5761-21	2102354VQK10QA106248
	HQ/1F/R08/W/12	Huawei AirEngine5761-21	2102354VQK10QA106094
	HQ/1F/R09/W/13	Huawei AirEngine5761-21	2102354VQK10QA106123
Main Building Second Floor	2F_RS_R10_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017714
	2F_SS_R11_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017785
	2F_SS_R12_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001376
	HQ/2F/R10/W/14	Huawei AirEngine5761-21	2102354VQK10QA106135
	HQ/2F/R10/W/15	Huawei AirEngine5761-21	2102354VQK10QA106225
	HQ/2F/R12/W/16	Huawei AirEngine5761-21	2102354VQK10QA106143
	HQ/2F/R10/W/17	Huawei AirEngine5761-21	2102354VQK10QA106144
	HQ/2F/R11/W/18	Huawei AirEngine5761-21	2102354VQK10QA106134

Main Building Third Floor	3F_SS_R13_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017803
	3F_RS_R14_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017762
	HQ/3F/R14/W/19	Huawei AirEngine5761-21	2102354VQK10QA106163
	HQ/3F/R14/W/20	Huawei AirEngine5761-21	2102354VQK10QA107003
	HQ/3F/R14/W/21	Huawei AirEngine5761-21	2102354VQK10QA106905
	HQ/3F/R13/W/22	Huawei AirEngine5761-21	2102354VQK10QA106008
	HQ/3F/R13/W/23	Huawei AirEngine5761-21	2102354VQK10QA106232
	HQ/3F/R13/W/24	Huawei AirEngine5761-21	2102354VQK10QA105994
	HQ/3F/R13/W/51	Huawei AirEngine5761-21	2102354VQK10QA106139
Main Building Fourth Floor	4F_RS_R15_SW1	Huawei CloudEngine L48T4S-A-V2 L3 Switch Non PoE	4E2490003878
	4F_RS_R16_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017682
	4F_SS_R17_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017679
	4F_SS_R18_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017663
	4F_SS_R19_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6000830
	HQ/4F/R16/W/25	Huawei AirEngine5761-21	2102354VQK10QA106231
	HQ/4F/R16/W/26	Huawei AirEngine5761-21	2102354VQK10QA106095
	HQ/4F/R16/W/27	Huawei AirEngine5761-21	2102354VQK10QA105964
	HQ/4F/R18/W/28	Huawei AirEngine5761-21	2102354VQK10QA106086
	HQ/4F/R17/W/29	Huawei AirEngine5761-21	2102354VQK10QA106154

Main Building Fifth Floor	5F_SS_R20_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001318
	5F_SS_R20_SW2	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001370
	5F_SS_R21_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001196
	5F_RS_R22_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	4E2490003731
	5F_RS_R22_SW2	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001246
	HQ/5F/R21/W/30	Huawei AirEngine5761-21	2102354VQK10QA107006
	HQ/5F/R20/W/31	Huawei AirEngine5761-21	2102354VQK10QA106908
	HQ/5F/R22/W/32	Huawei AirEngine5761-21	2102354VQK10QA107031
	HQ/5F/R22/W/33	Huawei AirEngine5761-21	2102354VQK10QA106914
	HQ/5F/R22/W/34	Huawei AirEngine5761-21	2102354VQK10QA106972
Main Building Sixth Floor	6F_RS_R23_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017734
	6F_RS_R24_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001477
	6F_SS_R25_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001362
	6F_SS_R26_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017644
	6F_SS_R26_SW2	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001377
	HQ/6F/R23/W/35	Huawei AirEngine5761-21	2102354VQK10QA107022
	HQ/6F/R23/W/36	Huawei AirEngine5761-21	2102354VQK10QA106066
	HQ/6F/R24/W/37	Huawei AirEngine5761-21	2102354VQK10QA106106
	HQ/6F/R26/W/38	Huawei AirEngine5761-21	2102354VQK10QA106033
	HQ/6F/R25/W/39	Huawei AirEngine5761-21	2102354VQK10QA106253
	HQ/6F/R25/W/40	Huawei AirEngine5761-21	2102354VQK10QA105952

Main Building Seventh Floor	7F_RS_R27_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017755
	7F_RS_R28_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017701
	7F_RS_R29_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001265
	7F_RS_R30_SW1	Huawei CloudEngine L48T4S-A-V2 L3 Switch Non PoE	4E2490003926
	7F_SS_R31_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6001113
	HQ/7F/R28/W/41	Huawei AirEngine5761-21	2102354VQK10QA106251
	HQ/7F/R27/W/42	Huawei AirEngine5761-21	2102354VQK10QA106034
	HQ/7F/R31/W/43	Huawei AirEngine5761-21	2102354VQK10QA106257
	HQ/7F/R31/W/44	Huawei AirEngine5761-21	2102354VQK10QA105969
	HQ/7F/R31/W/45	Huawei AirEngine5761-21	2102354VQK10QA106036
	HQ/7F/R31/W/46	Huawei AirEngine5761-21	2102354VQK10QA105968
	HQ/7F/R27/W/52	Huawei AirEngine5761-21	2102354VQK10QA105959
Canteen Building Ground Floor	CNTN_GF_R32_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6000795
	CAN/GF/R32/W/47	Huawei AirEngine5761-21	2102354VQK10QA106026
Canteen Building Second Floor	CNTN_2F_R33_SW1	Huawei CloudEngine L48LP4S-A-V2 L3 Switch PoE	QU2496017670
	CAN/2F/R33/W/48	Huawei AirEngine5761-21	2102354VQK10QA105942
	CAN/2F/R33/W/49	Huawei AirEngine5761-21	2102354VQK10QA105967
Canteen Building Third Floor	CNTN_3F_R34_SW1	Huawei CloudEngine L24P4S-A-V2 L3 Switch PoE	QU24A6000946
	CAN/3F/R34/W/50	Huawei AirEngine5761-21	2102354VQK10QA106045

6 IP Address Plan

6.1 Devices Management IP Addresses

Product	Management IP
Huawei iMaster NEC-Campus	10.220.20.112

Building Floor	Switch Host Name	Switch Management IP
Main Building Ground Floor	Core_1	10.220.20.253
	Core_2	10.220.20.254
	GF_SS_R1_SW1	10.220.20.2
	GF_SS_R2_SW1	10.220.20.3
	GF_RS_R3_SW1	10.220.20.4
	GF_RS_R4_SW1	10.220.20.5
Main Building First Floor	1F_RS_R5_SW1	10.220.20.11
	1F_RS_R6_SW1	10.220.20.12
	1F_RS_R7_SW1	10.220.20.13
	1F_SS_R8_SW1	10.220.20.14
	1F_SS_R9_SW1	10.220.20.15
Main Building Second Floor	2F_RS_R10_SW1	10.220.20.21
	2F_SS_R11_SW1	10.220.20.22
	2F_SS_R12_SW1	10.220.20.23
Main Building Third Floor	3F_SS_R13_SW1	10.220.20.31
	3F_RS_R14_SW1	10.220.20.32

Main Building Fourth Floor	4F_RS_R15_SW1	10.220.20.41
	4F_RS_R16_SW1	10.220.20.42
	4F_SS_R17_SW1	10.220.20.43
	4F_SS_R18_SW1	10.220.20.44
	4F_SS_R19_SW1	10.220.20.57
Main Building Fifth Floor	5F_SS_R20_SW1	10.220.20.51
	5F_SS_R20_SW2	10.220.20.52
	5F_SS_R21_SW1	10.220.20.54
	5F_RS_R22_SW1	10.220.20.55
	5F_RS_R22_SW2	10.220.20.56
Main Building Sixth Floor	6F_RS_R23_SW1	10.220.20.61
	6F_RS_R24_SW1	10.220.20.62
	6F_SS_R25_SW1	10.220.20.63
	6F_SS_R26_SW1	10.220.20.64
	6F_SS_R26_SW2	10.220.20.65
Main Building Seventh Floor	7F_RS_R27_SW1	10.220.20.71
	7F_RS_R28_SW1	10.220.20.72
	7F_RS_R29_SW1	10.220.20.73
	7F_RS_R30_SW1	10.220.20.74
	7F_SS_R31_SW1	10.220.20.75
Canteen Building Ground Floor	CNTN_GF_R32_SW1	10.220.20.81
Canteen Building Second Floor	CNTN_2F_R33_SW1	10.220.20.82
Canteen Building Third Floor	CNTN_3F_R34_SW1	10.220.20.83

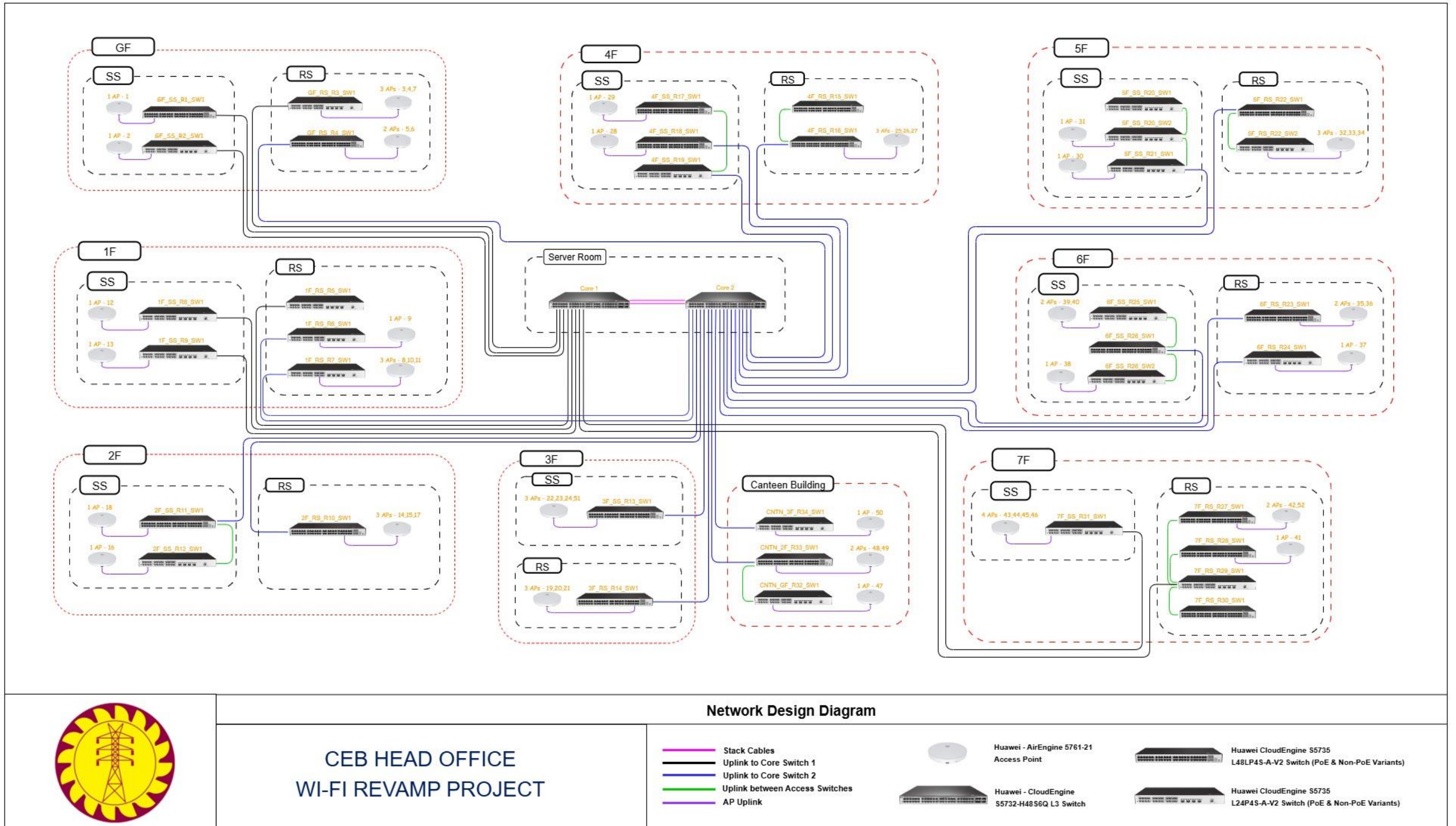
6.2 Network VLAN and Subnet Mapping

ID	NAME	ADD RANGE	GATEWAY	SUBNET MASK
500	DGM(IT)-Office	10.228.192.0	10.228.192.1	255.255.254.0
501	CE(ESNS)	10.228.194.0	10.228.194.1	255.255.254.0
502	ISD-Unit(IT)	10.228.196.0	10.228.196.1	255.255.254.0
503	ISO-Unit(IT)	10.228.198.0	10.228.198.1	255.255.254.0
504	S&A-Unit(IT)	10.228.200.0	10.228.200.1	255.255.254.0
505	BP-Unit(IT)	10.228.202.0	10.228.202.1	255.255.254.0
506	ESNS-ITOs	10.228.204.0	10.228.204.1	255.255.254.0
508	ESNS-Trainees	10.228.207.0	10.228.207.1	255.255.255.240
509	ISD-Trainees	10.228.207.112	10.228.207.113	255.255.255.240
510	BP-Trainees	10.228.207.128	10.228.207.129	255.255.255.240
511	ISO-Trainees	10.228.207.144	10.228.207.145	255.255.255.240
513	CE(Tx-Grid-Design)	10.237.2.0	10.237.2.1	255.255.254.0
514	CE(Tx-Line-Design)	10.237.4.0	10.237.4.1	255.255.254.0
515	Head-Office-Library	10.244.15.0	10.244.15.1	255.255.255.240
516	DGM(Trans-&-Generation-Planning)	10.236.32.0	10.236.32.1	255.255.254.0
517	CE(Tx-Planning)	10.236.34.0	10.236.34.1	255.255.254.0
518	CE(Generation-Planning)	10.236.36.0	10.236.36.1	255.255.254.0
519	CE(Generation-Development-Studies)	10.236.38.0	10.236.38.1	255.255.254.0
521	AGM(Tx-Wired)-Office	10.233.255.0	10.233.255.1	255.255.255.0
522	DGM(C&R-Relations)Tr-Office	10.233.128.0	10.233.128.1	255.255.255.0
523	HRO-C&R-Relations-Tr.-Wr.	10.233.129.0	10.233.129.1	255.255.255.0
524	EE-C&R-Relations-Tr-Wr	10.233.130.0	10.233.130.1	255.255.255.0
526	Chairman-Office	10.244.0.0	10.244.0.1	255.255.255.0

ID	NAME	ADD RANGE	GATEWAY	SUBNET MASK
528	GM-Office_ Executives	10.244.2.0	10.244.2.1	255.255.255.0
529	GM-Office_ Staff	10.244.1.0	10.244.1.1	255.255.255.0
530	DGM(Personnel)-Office	10.244.12.0	10.244.12.1	255.255.255.0
531	Chief-Engineer(Policy)	10.244.13.0	10.244.13.1	255.255.255.0
532	Welfare_ Manager	10.244.14.0	10.244.14.1	255.255.255.0
534	DGM-P&D-DD1	10.226.64.0	10.226.64.1	255.255.255.0
535	CE(P&D)-DD1	10.226.66.0	10.226.66.1	255.255.255.0
536	CE(MM&Procurement)-DD1	10.226.68.0	10.226.68.1	255.255.255.0
537	DGM(C&C)-DD1	10.226.32.0	10.226.32.1	255.255.255.0
539	AFM(Tx)-NWO	10.236.192.0	10.236.192.1	255.255.254.0
540	AGM(Tx)-NWO	10.237.255.0	10.237.255.1	255.255.255.0
545	Internal_Audit	10.244.4.0	10.244.4.1	255.255.255.0
546	DC	10.227.64.0	10.227.64.1	255.255.255.0
547	FM-SM&BSO	10.244.5.0	10.244.5.1	255.255.255.0
548	Special-Investigations	10.244.9.0	10.244.9.1	255.255.255.0
550	Training-Branch-Liaison-Office	10.235.5.0	10.235.5.1	255.255.255.240
551	Legal-Branch	10.244.6.0	10.244.6.1	255.255.255.0
552	Accountant(Cash)-DD1	10.227.8.0	10.227.8.1	255.255.255.0
557	AFM(DD1)	10.244.5.0	10.244.5.1	255.255.255.0
558	AGM(CS)	10.229.255.0	10.229.255.1	255.255.255.0
559	DGM(EP)	10.236.224.0	10.236.224.1	255.255.255.0
560	DGM(Tx-Design&Env)	10.237.0.0	10.237.0.1	255.255.255.0
595	DefaultVlan	10.244.10.0	10.244.10.1	255.255.254.0
596	CHAIRMAN-IOT	10.228.209.128	10.228.209.129	255.255.255.240
597	GM-IOT	10.228.209.112	10.228.209.113	255.255.255.240
598	HQ-IOT-DEVICE	10.228.208.0	10.228.208.1	255.255.255.0
599	IT-IOT	10.228.209.0	10.228.209.1	255.255.255.240

7 Network Design Diagram

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8 Switch Port Zone – Core Switch Cluster

8.1 Core Switch

Local Interface	Neighbor Device	Neighbor Interface
GE1/0/1	GF_SS_R1_SW1	GE1/0/48
GE1/0/5	1F_SS_R9_SW1	GE1/0/24
GE1/0/7	GF_SS_R2_SW1	GE1/0/24
GE1/0/22	1F_RS_R5_SW1	GE1/0/24
GE1/0/24	Core-Stack	GE2/0/24
GE1/0/25	7F_SS_R29_SW1	GE1/0/24
GE1/0/43	FortiGate-601F-Secondary	port3
GE1/0/44	FortiGate-601F-Primary	port4
GE2/0/1	1F_RS_R6_SW1	GE1/0/24
GE2/0/2	2F_RS_R10_SW1	GE1/0/48
GE2/0/3	3F_RS_R14_SW_1	GE1/0/48
GE2/0/5	5F_RS_R22_SW_1	GE1/0/48
GE2/0/6	6F_R23_RS_SW1	GE1/0/48
GE2/0/9	GF_RS_R4_SW1	GE1/0/26
GE2/0/10	GF_RS_R3_SW1	GE1/0/24
GE2/0/11	4F_SS_R19_SW1	GE1/0/24
GE2/0/12	2F_SS_R11_SW1	GE1/0/48
GE2/0/13	3F_SS_R13_SW_1	GE1/0/48
GE2/0/14	4F_SS_R18_SW1	GE1/0/48
GE2/0/15	5F_SS_R21_SW1	GE1/0/24
GE2/0/16	6F_RS_R26_SW1	GE1/0/48
GE2/0/19	CANTEEN_3F_R34_SW1	GE1/0/24
GE2/0/20	7F_RS_R31_SW1	GE1/0/24
GE2/0/22	CANTEEN_2F_R33_SW1	GE1/0/48
GE2/0/23	1F_RS_R7_SW1	GE1/0/24
GE2/0/24	Core-Stack	GE1/0/24
GE2/0/43	FortiGate-601F-Secondary	port4
GE2/0/44	FortiGate-601F-Primary	port3

9 Configuration Information

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9.1 Core Switch Configuration

```

!Software Version V200R022C00SPC500
#
sysname Core-Stack
#
dns resolve
dns server 10.128.1.12
dns server 10.128.3.12
#
vlan batch 2 to 30 32 to 35 40 to 43 52 to 55 60
110 120 200 500 to 506 509 to 511
vlan batch 513 to 519 521 to 524 526 528 to 529
531 to 532 534 to 540 542 545 to 548 550 to 552
590
vlan batch 595 to 599 1100
#
lnp disable
#
stp mode rstp
stp instance 0 priority 0
#
authentication-profile name
default_authen_profile
authentication-profile name
dot1x_authen_profile
authentication-profile name
dot1xmac_authen_profile

authentication-profile name mac_authen_profile
authentication-profile name multi_authen_profile
authentication-profile name
portal_authen_profile
#
lldp auto-vlan 1100 sensor ap
lldp auto-vlan tagged 500 to 501 503 to 507 509
to 511 514 to 527 531 539 to 545 547 to 600
sensor ap
#
http server-source all-interface
#
clock timezone huawei add 05:30:00
#
pnp startup-vlan send enable
pnp startup-vlan 1100
pnp wireless startup-vlan 1100
#

assign arp netconf number 200
#
undo hwtaacs enable
#
dhcp enable
#
dns snooping server-ip-address 10.128.1.12
dns snooping server-ip-address 10.128.3.12
#
diffserv domain default
#
radius-server template default
#
hwtaacs-server accounting-stop-packet resend
disable
#
pki realm default
certificate-check none
#
acl number 3002
rule 5 permit ip destination 10.135.100.0
0.0.0.255
rule 90 deny ip destination 10.0.0.0
0.255.255.255
rule 95 deny ip destination 172.16.0.0
0.15.255.255
rule 100 deny ip destination 192.168.0.0
0.0.255.255
acl number 3012
rule 2 permit ip destination 10.128.20.112 0
rule 3 permit ip destination 10.128.1.12 0
rule 4 permit ip destination 10.128.3.12 0
rule 5 deny ip destination 10.0.0.0 0.255.255.255
acl number 3333
rule 5 permit icmp source 10.220.20.5 0
destination 10.128.20.112 0
rule 10 permit icmp source 10.128.20.112 0
destination 10.220.20.5 0
#
qos-profile name CEB-Guest_Mid-Permission
car cir 5000 pir 5000 cbs 940000 pbs 1565000
inbound
car cir 5000 pir 5000 cbs 940000 pbs 1565000
outbound
qos-profile name CEB-Guest_General-Permission
car cir 1000 pir 1000 cbs 188000 pbs 313000
inbound

```

```
car cir 1000 pir 1000 cbs 188000 pbs 313000      name vlan22
outbound                                           vlan 23
#                                                  name vlan23
free-rule-template name default_free_rule         vlan 24
#                                                  name vlan24
portal-access-profile name portal_access_profile  vlan 25
#                                                  name vlan25
drop-profile default                             vlan 26
#                                                  name vlan26
vlan 1                                             vlan 27
  name vlan1                                       name vlan27
vlan 2                                             vlan 28
  name vlan2                                       name vlan28
vlan 3                                             vlan 29
  name vlan3                                       name vlan29
vlan 4                                             vlan 30
  name vlan4                                       name vlan30
vlan 5                                             vlan 32
  name vlan5                                       name vlan32
vlan 6                                             vlan 33
  name vlan6                                       name vlan33
vlan 7                                             vlan 34
  name vlan7                                       name vlan34
vlan 8                                             vlan 35
  name vlan8                                       name vlan35
vlan 9                                             vlan 40
  name vlan9                                       name vlan40
vlan 10                                           vlan 41
  name vlan10                                      name vlan41
vlan 11                                           vlan 42
  name vlan11                                      name vlan42
vlan 12                                           vlan 43
  name vlan12                                      name vlan43
vlan 13                                           vlan 52
  name vlan13                                      name vlan52
vlan 14                                           vlan 53
  name vlan14                                      name vlan53
vlan 15                                           vlan 54
  name vlan15                                      name vlan54
vlan 16                                           vlan 55
  name vlan16                                      name vlan55
vlan 17                                           vlan 60
  name vlan17                                      name vlan60
vlan 18                                           vlan 110
  name vlan18                                      name vlan110
vlan 19                                           vlan 120
  name vlan19                                      name vlan120
vlan 20                                           vlan 200
  name vlan20                                      name vlan200
vlan 21                                           vlan 500
  name vlan21                                      name vlan500
vlan 22                                           vlan 501
```

name vlan501	name vlan535
vlan 502	vlan 536
name vlan502	name vlan536
vlan 503	vlan 537
name vlan503	name vlan537
vlan 504	vlan 538
name vlan504	name vlan538
vlan 505	vlan 539
name vlan505	name vlan539
vlan 506	vlan 540
name vlan506	name vlan540
vlan 509	vlan 542
name vlan509	name vlan542
vlan 510	vlan 545
name vlan510	name vlan545
vlan 511	vlan 546
name vlan511	name vlan546
vlan 513	vlan 547
name vlan513	name vlan547
vlan 514	vlan 548
name vlan514	name vlan548
vlan 515	vlan 550
name vlan515	name vlan550
vlan 516	vlan 551
name vlan516	name vlan551
vlan 517	vlan 552
name vlan517	name vlan552
vlan 518	vlan 590
name vlan518	name vlan590
vlan 519	vlan 595
name vlan519	name vlan595
vlan 521	vlan 596
name vlan521	name vlan596
vlan 522	vlan 597
name vlan522	name vlan597
vlan 523	vlan 598
name vlan523	name vlan598
vlan 524	vlan 599
name vlan524	name vlan599
vlan 526	vlan 1100
name vlan526	description ##MGMT##
vlan 528	name vlan1100
name vlan528	#
vlan 529	aaa
name vlan529	authentication-scheme default
vlan 531	authentication-mode local
name vlan531	authentication-scheme radius
vlan 532	authentication-mode radius
name vlan532	authorization-scheme default
vlan 534	authorization-mode local
name vlan534	accounting-scheme default
vlan 535	accounting-mode none

```

local-aaa-user password policy administrator #
password history record number 0 interface Vlanif3
password expire 0 ip address 10.135.15.1 255.255.255.0
domain default dhcp select relay
authentication-scheme radius dhcp relay server-ip 10.128.1.191
accounting-scheme default dhcp relay server-ip 10.128.1.192
radius-server default #
domain default_admin interface Vlanif4
authentication-scheme default ip address 10.135.20.1 255.255.255.0
accounting-scheme default dhcp select relay
domain domain_accampus dhcp relay server-ip 10.128.1.191
authentication-scheme default dhcp relay server-ip 10.128.1.192
accounting-scheme default #
radius-server default interface Vlanif5
local-user admin password irreversible-cipher ip address 10.135.25.1 255.255.255.0
$1c$,JlIN3),8>$vRF"8t"+yQ'~V(3vS_p93uhjVzt,Q9 dhcp select relay
@,kWH@*#`:$ dhcp relay server-ip 10.128.1.191
local-user admin privilege level 15 dhcp relay server-ip 10.128.1.192
local-user admin service-type terminal ssh http #
local-user Observer password irreversible-cipher interface Vlanif6
$1c$@75wAup%LE$I5v5&ZjES<&wa@*:NG&Vclf/ ip address 10.135.30.1 255.255.255.0
;_(~EH}FCK<6lKv6$ dhcp select interface
local-user Observer privilege level 1 dhcp server dns-list 10.128.1.12 10.128.3.12
local-user Observer service-type ssh http #
local-user cebadmin password irreversible-cipher interface Vlanif7
$1c$Sw.[J4x4$V$m[[Bgx[>4gWP"/c8M.>rZrcl'1z; ip address 10.135.35.1 255.255.255.0
Z~XSe!zu#|U$ dhcp select relay
local-user cebadmin privilege level 15 dhcp relay server-ip 10.128.1.191
local-user cebadmin service-type terminal ssh dhcp relay server-ip 10.128.1.192
http #
local-user vsisadmin password irreversible-cipher interface Vlanif8
$1c$BQUHUP3lC3$F*"CD=z{,SHfks)/ul5H%GYtP}$ ip address 10.135.40.1 255.255.255.0
Q()Xk2C6b<iF~$ dhcp select relay
local-user vsisadmin privilege level 15 dhcp relay server-ip 10.128.1.191
local-user vsisadmin service-type terminal ssh dhcp relay server-ip 10.128.1.192
http #
local-user accampus@domain_accampus interface Vlanif9
password irreversible-cipher ip address 10.135.45.1 255.255.255.0
$1c$*h.WL>'*"C$jgV.Gi[bcC.*BW>^.i++cdaOly- dhcp select interface
/4TUgRoLErUgS$ dhcp server dns-list 10.128.1.12
local-user accampus@domain_accampus #
privilege level 15 interface Vlanif10
local-user accampus@domain_accampus service- ip address 10.135.50.1 255.255.255.0
type ssh http dhcp select interface
# dhcp server dns-list 10.128.1.12 10.128.3.12
interface Vlanif1 #
# interface Vlanif11
interface Vlanif2 ip address 10.135.55.1 255.255.255.0
ip address 10.135.10.1 255.255.255.0 dhcp select relay
dhcp select relay dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.191 dhcp relay server-ip 10.128.1.192
dhcp relay server-ip 10.128.1.192 #

```

```
interface Vlanif12
ip address 10.135.60.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif13
ip address 10.135.65.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif14
ip address 10.135.66.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif15
ip address 10.135.70.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif16
ip address 10.135.75.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif17
ip address 10.135.76.1 255.255.255.0
arp-proxy inner-sub-vlan-proxy enable
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif18
ip address 10.135.100.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif19
ip address 10.135.101.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif20
ip address 10.136.10.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
```

```
interface Vlanif21
ip address 10.136.15.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif22
ip address 10.135.11.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif23
ip address 10.130.8.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif24
ip address 10.130.1.1 255.255.255.0
arp-proxy enable
arp-proxy inner-sub-vlan-proxy enable
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif25
ip address 10.130.2.1 255.255.255.0
arp-proxy enable
arp-proxy inner-sub-vlan-proxy enable
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif26
ip address 10.130.3.1 255.255.255.0
arp-proxy enable
arp-proxy inner-sub-vlan-proxy enable
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif27
ip address 10.130.4.1 255.255.255.0
arp-proxy enable
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif28
ip address 10.130.5.1 255.255.255.0
dhcp select interface
```

```
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif29
ip address 10.130.7.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif30
ip address 10.130.30.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif32
ip address 10.130.100.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif33
ip address 10.162.0.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif34
ip address 10.162.2.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif35
ip address 10.162.3.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif40
ip address 10.137.10.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif41
ip address 10.137.20.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif42
ip address 10.137.15.1 255.255.255.0

dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif43
ip address 10.130.6.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif52
ip address 10.142.10.4 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif53
ip address 10.135.110.1 255.255.255.0
ip address 10.135.110.9 255.255.255.0 sub
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif54
ip address 10.135.125.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif55
ip address 10.135.126.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif60
ip address 10.135.80.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif110
ip address 10.0.0.1 255.255.0.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif120
ip address 10.135.120.1 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
```

```
interface Vlanif200
ip address 10.143.10.3 255.255.255.0
dhcp select relay
dhcp relay server-ip 10.128.1.191
dhcp relay server-ip 10.128.1.192
#
interface Vlanif500
ip address 10.228.192.1 255.255.254.0
dhcp select interface
dhcp server excluded-ip-address 10.228.192.2
10.228.192.255
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif501
ip address 10.228.194.1 255.255.254.0
dhcp select interface
dhcp server excluded-ip-address 10.228.194.2
10.228.194.255
dhcp server static-bind ip-address 10.228.195.35
mac-address 701a-b8c5-9d92
dhcp server static-bind ip-address 10.228.195.67
mac-address c4bd-e5ab-9b31
dhcp server static-bind ip-address 10.228.195.99
mac-address fe3e-ac41-2e5d
dhcp server static-bind ip-address
10.228.195.111 mac-address 06ad-74d2-e894
dhcp server static-bind ip-address
10.228.195.112 mac-address c4bd-e5ab-cb06
dhcp server static-bind ip-address
10.228.195.120 mac-address c4bd-e5ab-df3d
dhcp server static-bind ip-address
10.228.195.121 mac-address 6c24-0882-d13a
dhcp server static-bind ip-address
10.228.195.135 mac-address 38f3-ab19-da43
dhcp server static-bind ip-address
10.228.195.136 mac-address d03c-1f6b-65fc
dhcp server static-bind ip-address
10.228.195.137 mac-address d4be-d9bd-7c8f
dhcp server static-bind ip-address
10.228.195.140 mac-address a46b-b6bc-c5a0
dhcp server static-bind ip-address
10.228.195.141 mac-address d03c-1f6b-426b
dhcp server static-bind ip-address
10.228.195.142 mac-address 38f3-ab19-d289
dhcp server static-bind ip-address
10.228.195.143 mac-address 0456-e54d-305a
dhcp server static-bind ip-address
10.228.195.157 mac-address c4bd-e5ab-b779
dhcp server static-bind ip-address
10.228.195.158 mac-address 6c24-0882-d018
dhcp server dns-list 10.128.3.12 10.128.1.12

dhcp server domain-name ceb.local
#
interface Vlanif502
ip address 10.228.196.1 255.255.254.0
dhcp select interface
dhcp server excluded-ip-address 10.228.196.2
10.228.196.255
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif503
ip address 10.228.198.1 255.255.254.0
dhcp select interface
dhcp server dns-list 10.228.198.1
dhcp server domain-name ceb.local
#
interface Vlanif504
ip address 10.228.200.1 255.255.254.0
dhcp select interface
dhcp server excluded-ip-address 10.228.200.2
10.228.200.255
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif505
ip address 10.228.202.1 255.255.254.0
dhcp select interface
dhcp server excluded-ip-address 10.228.202.2
10.228.202.255
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif506
ip address 10.228.204.1 255.255.254.0
dhcp select interface
dhcp server excluded-ip-address 10.228.204.2
10.228.204.255
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif509
ip address 10.228.207.1 255.255.255.224
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif510
ip address 10.228.207.129 255.255.255.240
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif511
ip address 10.228.207.145 255.255.255.240
dhcp select interface
```

```
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif513
ip address 10.237.2.1 255.255.254.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif514
ip address 10.237.4.1 255.255.254.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif515
ip address 10.244.15.1 255.255.255.240
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif516
ip address 10.236.32.1 255.255.254.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif517
ip address 10.236.34.1 255.255.254.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif518
ip address 10.236.36.1 255.255.254.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif519
ip address 10.236.38.1 255.255.254.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif521
ip address 10.233.255.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif522
ip address 10.233.128.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif523
ip address 10.233.129.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
dhcp select interface
```

```
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif524
ip address 10.233.130.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif526
ip address 10.244.0.1 255.255.255.0
dhcp select interface
dhcp server static-bind ip-address 10.244.0.161
mac-address a46b-b62e-e131
dhcp server static-bind ip-address 10.244.0.219
mac-address 6c62-6d0d-4851
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif528
ip address 10.244.1.1 255.255.255.128
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif529
ip address 10.244.1.129 255.255.255.128
dhcp select interface
dhcp server static-bind ip-address 10.244.1.130
mac-address 70b5-e826-6cc0
dhcp server static-bind ip-address 10.244.1.131
mac-address 3497-f69d-20c9
dhcp server static-bind ip-address 10.244.1.132
mac-address 18a9-05ed-8009
dhcp server static-bind ip-address 10.244.1.133
mac-address dc4a-3e46-6ce7
dhcp server static-bind ip-address 10.244.1.134
mac-address 7412-b3d6-c63b
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif531
ip address 10.244.13.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.3.12 10.128.1.12
#
interface Vlanif532
ip address 10.244.14.1 255.255.255.0
dhcp select interface
dhcp server domain-name ceb.local
#
interface Vlanif534
ip address 10.226.64.1 255.255.255.0
dhcp select interface
```

```
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif535
ip address 10.226.66.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif536
ip address 10.226.68.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif537
ip address 10.226.32.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif538
ip address 10.244.12.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif539
ip address 10.236.192.1 255.255.255.0
dhcp select interface
dhcp server static-bind ip-address 10.236.192.2
mac-address b8ae-edb5-a24e
dhcp server static-bind ip-address 10.236.192.3
mac-address e454-e8c1-28b6
dhcp server static-bind ip-address 10.236.192.4
mac-address e454-e8c1-2a77
dhcp server static-bind ip-address 10.236.192.5
mac-address 2088-10a0-91b3
dhcp server static-bind ip-address 10.236.192.6
mac-address f48e-3896-8d5f
dhcp server static-bind ip-address 10.236.192.7
mac-address 5cea-1d2c-11c9
dhcp server static-bind ip-address 10.236.192.8
mac-address 8cec-4b5a-39d4
dhcp server static-bind ip-address 10.236.192.9
mac-address 441c-a840-3e31
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif540
ip address 10.237.255.1 255.255.255.0
dhcp select interface
```

```
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif542
ip address 10.244.8.1 255.255.255.0
arp-proxy inner-sub-vlan-proxy enable
dhcp select interface
dhcp server static-bind ip-address 10.244.8.2
mac-address b8ac-6fbc-dca7
dhcp server static-bind ip-address 10.244.8.3
mac-address 94c6-91f9-0b1d
dhcp server static-bind ip-address 10.244.8.4
mac-address b00c-d19e-029a
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif545
ip address 10.244.4.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif546
ip address 10.227.64.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif547
ip address 10.244.5.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif548
ip address 10.244.9.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif550
ip address 10.235.5.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif551
ip address 10.244.6.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif552
ip address 10.227.8.1 255.255.255.0
dhcp select interface
```

```

dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server domain-name ceb.local
#
interface Vlanif590
ip address 10.228.210.1 255.255.255.0
dhcp select interface
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif595
ip address 10.244.10.1 255.255.254.0
arp-proxy inner-sub-vlan-proxy enable
dhcp select interface
dhcp server static-bind ip-address 10.244.10.50
mac-address c03f-d5f5-d421
dhcp server static-bind ip-address 10.244.10.78
mac-address 0c84-dc7c-fab7
dhcp server static-bind ip-address 10.244.10.107
mac-address 84a9-3e5f-c1a0
dhcp server static-bind ip-address 10.244.10.137
mac-address b083-fe75-c4f0
dhcp server static-bind ip-address 10.244.10.171
mac-address b00c-d19e-029a
dhcp server static-bind ip-address 10.244.10.230
mac-address 0442-1a2a-48df
dhcp server static-bind ip-address 10.244.11.59
mac-address 646e-e0cb-139e
dhcp server static-bind ip-address 10.244.11.111
mac-address 00e0-4c40-d9a0
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif596
ip address 10.228.209.129 255.255.255.240
dhcp select interface
dhcp server dns-list 10.128.1.12 8.8.8.8
#
interface Vlanif597
ip address 10.228.209.113 255.255.255.240
dhcp select interface
dhcp server lease unlimited
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif598
ip address 10.228.208.1 255.255.255.0
dhcp select interface
dhcp server excluded-ip-address 10.228.208.3
10.228.208.38
dhcp server excluded-ip-address 10.228.208.61
dhcp server excluded-ip-address 10.228.208.100
10.228.208.110
dhcp server excluded-ip-address 10.228.208.122
10.228.208.123

dhcp server static-bind ip-address 10.228.208.2
mac-address 349f-7ba5-e5b2
dhcp server static-bind ip-address 10.228.208.39
mac-address 0080-91f1-b626
dhcp server static-bind ip-address 10.228.208.83
mac-address 0080-91d0-4549
dhcp server static-bind ip-address 10.228.208.86
mac-address 74bf-c014-e090
dhcp server static-bind ip-address
10.228.208.187 mac-address 5838-7915-5dc1
dhcp server static-bind ip-address
10.228.208.188 mac-address 7438-b750-d7fa
dhcp server static-bind ip-address
10.228.208.192 mac-address 349f-7ba1-119d
dhcp server static-bind ip-address
10.228.208.194 mac-address 349f-7ba1-10cf
dhcp server static-bind ip-address
10.228.208.213 mac-address 0080-91cc-c0e3
dhcp server static-bind ip-address
10.228.208.246 mac-address 74bf-c015-5db7
dhcp server static-bind ip-address
10.228.208.253 mac-address 1060-4b86-b333
dhcp server lease unlimited
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif599
ip address 10.228.209.1 255.255.255.240
dhcp select interface
dhcp server excluded-ip-address 10.228.209.2
10.228.209.8
dhcp server lease unlimited
dhcp server dns-list 10.128.1.12 10.128.3.12
#
interface Vlanif1100
ip address 10.220.20.254 255.255.255.0
dhcp select interface
dhcp server excluded-ip-address 10.220.20.1
10.220.20.100
dhcp server dns-list 10.128.1.12 10.128.3.12
dhcp server option 148 ascii agilemode=agile-
cloud;agilemanage-mode=ip;agilemanage-
domain=10.128.20.112;agilemanage-
port=10020;sitecode=618941ab42e71717;
#
interface MEth0/0/1
ip address 192.168.1.253 255.255.255.0
#
interface MEth0/0/2
ip address 169.254.2.1 255.255.255.252
#
interface Eth-Trunk52
description ##Fortigate Firewall 01##

```

```
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 52
trust dscp
#
interface Eth-Trunk53
description ##Fortigate Firewall 02##
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 52
trust dscp
#
interface GigabitEthernet1/0/1
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/2
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/3
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/4
port link-type access
port default vlan 25
trust dscp
#
interface GigabitEthernet1/0/5
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/6
port link-type access
port default vlan 25
trust dscp
#
interface GigabitEthernet1/0/7
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/8
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
```

```
interface GigabitEthernet1/0/9
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/10
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/11
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/12
port link-type access
port default vlan 110
trust dscp
#
interface GigabitEthernet1/0/13
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/14
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/15
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/16
port link-type access
port default vlan 25
trust dscp
#
interface GigabitEthernet1/0/17
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/18
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/19
port link-type trunk
```

```
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 53 110 1100
trust dscp
#
interface GigabitEthernet1/0/20
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/21
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/22
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/23
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/24
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
mad detect mode direct
#
interface GigabitEthernet1/0/25
description 7F-RS-Call-Center
port link-type trunk
port trunk pvid vlan 16
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/26
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/27
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/28
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
```

```
interface GigabitEthernet1/0/29
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/30
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/31
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/32
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/33
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/34
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/35
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/36
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/37
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/38
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/39
port link-type trunk
```

```
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/40
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/41
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/42
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet1/0/43
eth-trunk 52
#
interface GigabitEthernet1/0/44
eth-trunk 53
#
interface GigabitEthernet2/0/1
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/2
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/3
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/4
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/5
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/6
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/7
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/8
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/9
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/10
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/11
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/12
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/13
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/14
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/15
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/16
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
```

```
interface GigabitEthernet2/0/17
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/18
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/19
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/20
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/21
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/22
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/23
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/24
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
mad detect mode direct
#
interface GigabitEthernet2/0/25
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/26
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/27
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/28
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/29
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/30
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/31
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/32
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/33
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/34
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/35
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/36
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/37
port link-type trunk
port trunk allow-pass vlan 2 to 4094
```

```
trust dscp
#
interface GigabitEthernet2/0/38
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/39
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/40
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/41
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/42
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface GigabitEthernet2/0/43
eth-trunk 52
#
interface GigabitEthernet2/0/44
eth-trunk 53
#
interface XGigabitEthernet1/0/1
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface XGigabitEthernet1/0/2
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface XGigabitEthernet1/0/3
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface XGigabitEthernet1/0/4
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp

#
interface XGigabitEthernet2/0/1
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface XGigabitEthernet2/0/2
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface XGigabitEthernet2/0/3
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface XGigabitEthernet2/0/4
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface 40GE1/0/1
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface 40GE1/0/2
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface 40GE1/0/3
#
interface 40GE1/0/4
#
interface 40GE1/0/5
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface 40GE1/0/6
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface 40GE2/0/1
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface 40GE2/0/2
port link-type trunk
```

```
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface 40GE2/0/3
#
interface 40GE2/0/4
#
interface 40GE2/0/5
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface 40GE2/0/6
port link-type trunk
port trunk allow-pass vlan 2 to 4094
trust dscp
#
interface NULL0
#
ospf 1 router-id 10.220.20.254
area 0.0.0.0
network 10.236.192.0 0.0.1.255
#
undo icmp name timestamp-request receive
#
ip route-static 0.0.0.0 0.0.0.0 10.142.10.3
ip route-static 10.135.110.0 255.255.255.0
10.0.0.21
ip route-static 10.150.60.0 255.255.255.0
10.135.75.2
ip route-static 10.150.61.0 255.255.255.0
10.135.75.2
ip route-static 192.168.0.0 255.255.255.0
10.0.0.21
#
stelnet server enable
ssh client first-time enable
ssh server-source all-interface
ssh ipv6 server-source all-interface
ssh server cipher aes256_ctr aes128_ctr
ssh server hmac sha2_256
ssh server key-exchange dh_group16_sha512
dh_group15_sha512 dh_group14_sha256
dh_group_exchange_sha256
ssh client cipher aes256_ctr aes128_ctr
ssh client hmac sha2_256
ssh client key-exchange dh_group16_sha512
dh_group15_sha512 dh_group14_sha256
dh_group_exchange_sha256
ssh server dh-exchange min-len 2048

ssh server publickey rsa_sha2_512 rsa_sha2_256
ecc
#
user-interface con 0
authentication-mode aaa
user-interface vty 0 4
authentication-mode aaa
protocol inbound all
user-interface vty 16 20
#
wlan
traffic-profile name default
security-profile name default
security-profile name default-wds
security-profile name default-mesh
ssid-profile name default
vap-profile name default
wds-profile name default
mesh-handover-profile name default
mesh-profile name default
regulatory-domain-profile name default
air-scan-profile name default
rrm-profile name default
radio-2g-profile name default
radio-5g-profile name default
wids-spoof-profile name default
wids-whitelist-profile name default
wids-profile name default
ap-system-profile name default
port-link-profile name default
wired-port-profile name default
ap auth-mode sn-auth
ap-group name default
provision-ap
#
dot1x-access-profile name dot1x_access_profile
#
mac-access-profile name mac_access_profile
#
netconf
callhome imaster-campus
ip address 10.128.20.112 port 10020
source ip interface Vlanif1100
#
ops
#
remote-unit
#
return
```

9.2 Access Switch Configuration

```
<3F_SS_R13_SW_1>dis cu
!Software Version V600R024C00SPC500
!Last configuration was updated at 2025-07-23 15:25:53+05:30 by huawei
!Last configuration was saved at 2025-07-23 16:56:43+05:30
#
pki realm default
#
clock timezone LKA add 05:30:00
#
sysname 3F_SS_R13_SW_1
#
ip route 10.0.0.0 255.0.0.0 Vlanif6 dhcp
#
undo ftp server source all-interface
undo ftp ipv6 server source all-interface
#
ssl policy default
pki-domain default
ssl minimum version tls1.2
cipher-suite exclude key-exchange rsa
cipher-suite exclude cipher mode cbc
cipher-suite exclude hmac sha1
diffie-hellman modulus 3072
ecdh group curve brainpool
signature algorithm-list ed25519 ed448 rsa-pss-pss-sha256 rsa-pss-pss-sha384 rsa-pss-pss-sha512 rsa-pss-rsae-sha256 rsa-pss-rsae-sha384 rsa-pss-rsae-sha512
#
info-center logfile compression lzma
#
device board 1 board-type S5735-L48LP4S-A-V2
#
dns resolve
#
authentication-profile name
Auth42f3b387e927016d4df7
dot1x-access-profile
Dot1x42f3b387e927016d4df7
```

```
access-domain aaa42f3b387e927016d4df7
force
authentication event authen-fail action
authorize service-scheme
AuthFail42f3b387e927016d4df7
authentication-profile name
default_authen_profile
authentication-profile name
dot1x_authen_profile
authentication-profile name
dot1xmac_authen_profile
authentication-profile name
mac_authen_profile
authentication-profile name
multi_authen_profile
authentication-profile name
portal_authen_profile
#
access-user dot1x-identity speed-limit 10
#
dns server 10.128.1.12
dns server 10.128.3.12
#
drop-profile default
#
ntp ipv6 server disable
ntp server source-interface all disable
ntp ipv6 server source-interface all disable
ntp unicast-server 10.128.1.12
#
vlan batch 6 25 27 54 to 55 110 125 500 to 600 1100 1200 to 1202 1204
#
stp mode rstp
#
lnp disable
#
error-down auto-recovery cause link-flap
interval 60
#
undo telnet server-source all-interface
undo telnet ipv6 server-source all-interface
```

```
# name vlan501
qos-profile CEB-Guest_Mid-Permission #
car cir 5000 pir 5000 cbs 940000 pbs 1565000 vlan 502
inbound name vlan502
car cir 5000 pir 5000 cbs 940000 pbs 1565000 #
outbound vlan 503
# name vlan503
qos-profile CEB-Guest_General-Permission #
car cir 1000 pir 1000 cbs 188000 pbs 313000 vlan 504
inbound name vlan504
car cir 1000 pir 1000 cbs 188000 pbs 313000 #
outbound vlan 505
# name vlan505
qos schedule-profile default #
# vlan 506
diffserv domain default name vlan506
# #
ip vpn-instance _management_vpn_ vlan 507
ipv4-family name vlan507
# #
vlan 1 vlan 508
name vlan1 name vlan508
# #
vlan 6 vlan 509
name vlan6 name vlan509
# #
vlan 25 vlan 510
# name vlan510
vlan 27 #
# vlan 511
vlan 54 name vlan511
# #
vlan 55 vlan 512
description GM_Conference name vlan512
name vlan55 #
# vlan 513
vlan 110 name vlan513
name vlan110 #
# vlan 514
vlan 125 name vlan514
# #
vlan 500 vlan 515
name vlan500 name vlan515
# #
vlan 501 vlan 516
```

name vlan516	name vlan531
#	#
vlan 517	vlan 532
name vlan517	name vlan532
#	#
vlan 518	vlan 533
name vlan518	name vlan533
#	#
vlan 519	vlan 534
name vlan519	name vlan534
#	#
vlan 520	vlan 535
name vlan520	name vlan535
#	#
vlan 521	vlan 536
name vlan521	name vlan536
#	#
vlan 522	vlan 537
name vlan522	name vlan537
#	#
vlan 523	vlan 538
name vlan523	name vlan538
#	#
vlan 524	vlan 539
name vlan524	name vlan539
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vlan 525	vlan 540
name vlan525	name vlan540
#	#
vlan 526	vlan 541
name vlan526	name vlan541
#	#
vlan 527	vlan 542
name vlan527	name vlan542
#	#
vlan 528	vlan 543
name vlan528	name vlan543
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vlan 529	vlan 544
name vlan529	name vlan544
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vlan 530	vlan 545
name vlan530	name vlan545
#	#
vlan 531	vlan 546

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name vlan546
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vlan 547
name vlan547
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vlan 548
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vlan 558
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vlan 561
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```
name vlan561
#
vlan 562
name vlan562
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vlan 563
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vlan 564
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vlan 573
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vlan 575
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vlan 576
```

```
name vlan576
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vlan 577
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vlan 578
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vlan 579
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vlan 580
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vlan 581
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vlan 582
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vlan 588
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vlan 589
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vlan 590
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#
vlan 591
```

```
name vlan591
#
vlan 592
name vlan592
#
vlan 593
name vlan593
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vlan 594
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vlan 595
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vlan 596
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vlan 597
name vlan597
#
vlan 598
name vlan598
#
vlan 599
name vlan599
#
vlan 600
name vlan600
#
vlan 1100
name vlan1100
#
vlan 1204
#
acl number 3002
rule 5 name 5 permit ip destination
10.135.100.0 0.0.0.255
rule 90 name 90 deny ip destination 10.0.0.0
0.255.255.255
rule 95 name 95 deny ip destination
172.16.0.0 0.15.255.255
rule 100 name 100 deny ip destination
192.168.0.0 0.0.255.255
#
acl number 3012
```

```

rule 2 name 2 permit ip destination
10.128.20.112 0
rule 3 name 3 permit ip destination
10.128.1.12 0
rule 4 name 4 permit ip destination
10.128.3.12 0
rule 5 name 5 deny ip destination 10.0.0.0
0.255.255.255
#
aaa
authentication-scheme
ausch42f3b387e927016d4df7
authentication-mode radius
authentication-scheme default
authentication-mode local
authentication-scheme radius
authentication-mode radius
authorization-scheme default
authorization-mode local
accounting-scheme
acsch42f3b387e927016d4df7
accounting-mode radius
accounting realtime 15
accounting-scheme default
accounting-mode none
service-scheme
AuthFail42f3b387e927016d4df7
user-vlan 1
local-aaa-user password policy administrator
password history record number 0
password expire 0
password complexity two-of-kinds
domain aaa42f3b387e927016d4df7
authentication-scheme
ausch42f3b387e927016d4df7
accounting-scheme
acsch42f3b387e927016d4df7
statistic enable
radius-server radius4ec3b4d3e2f0b39756ab
domain default
authentication-scheme default
accounting-scheme default
domain default_admin
authentication-scheme default
accounting-scheme default

```

```

domain domain_accampus
authentication-scheme default
accounting-scheme default
local-user Observer password irreversible-
cipher $1d$,KV(O>oq^LghRh"~$.y7S&4nn-
Z+e_)&'lVvH#kzOS=It~#v1=.b$7C$
local-user Observer password-force-change
disable
local-user Observer privilege level 1
local-user Observer service-type ssh http
local-user accampus@domain_accampus
password irreversible-cipher
$1d$9Cx1>1HLrN*VKKT%$[M.gR5yD1/;\b+~n(
xw#6+T^Tt*5;l={Q,l\".}.1$
local-user accampus@domain_accampus
password-force-change disable
local-user accampus@domain_accampus
privilege level 3
local-user accampus@domain_accampus
service-type ssh http
local-user administrator password
irreversible-cipher
$1d$*\r%LhNkSRFHRXSA$s+$')svkE:4NKg+(vx
1KB%H}!Xd6p.rCC*Rfr/R:$
local-user administrator password-force-
change disable
local-user administrator privilege level 3
local-user administrator service-type terminal
ssh http
local-user cebadmin password irreversible-
cipher $1d$JHI-
81&ZT>7fSG%l$,x!s5sk,rl}>Cx!;8G7/3~IC&V$B
G=,DD->ytZu<$
local-user cebadmin password-force-change
disable
local-user cebadmin privilege level 3
local-user cebadmin service-type terminal ssh
http
local-user vsisadmin password irreversible-
cipher $1d$)dE#8J.-
=!!"874;$39]x@+5sA:cboU#9>@N~e5}q$N\'2
>NHXd&iZ=8K$
local-user vsisadmin password-force-change
disable
local-user vsisadmin privilege level 3

```

```
local-user vsisadmin service-type terminal ssh
http
#
radius-server template
radius4ec3b4d3e2f0b39756ab
radius-server shared-key cipher
%+%##!!!!!!"!!!!"!!!!*!!!!7LTnVOK_g:mHtU6
R{#CD*q9~#p.(DN9<+A~!!!!2jp5!!!!<!!!!|ij\
3nph8:_5p7=-]59Qr[5LV=Ub[,OHik>!!!!%+%#
radius-server accounting 10.128.20.112 1813
weight 80
radius-server authentication 10.128.20.112
1812 weight 80
radius-server authorization 10.128.20.112
shared-key cipher
%@%##!!!!!!"!!!!"!!!!*!!!!7LTnVOK_g:K$OG
UI>{U5ol/C"ue(=_qI\w%!!!!2jp5!!!!<!!!!6G)r
K4+o}M2VJX'6]5^Ny^izYr]!3:`Ts4J!!!!%@%#
radius local-ip any
#
free-rule-template name default_free_rule
#
portal-access-profile name
portal_access_profile
#
dot1x-access-profile name
Dot1x42f3b387e927016d4df7
dot1x-access-profile name
dot1x_access_profile
#
mac-access-profile name mac_access_profile
#
remote-unit
#
stack
#
license
#
warranty
#
interface Vlanif6
shutdown
ip address 10.135.30.245 255.255.255.0
#
interface Vlanif25
```

```
shutdown
ip address 10.130.2.250 255.255.255.0
#
interface Vlanif54
shutdown
ip address 10.135.125.231 255.255.255.0
#
interface Vlanif55
shutdown
ip address 10.135.126.2 255.255.255.0
#
interface Vlanif110
shutdown
ip address 10.0.0.130 255.255.0.0
#
interface Vlanif125
shutdown
#
interface Vlanif1100
ip address 10.220.20.31 255.255.255.0
#
interface GE1/0/1
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/2
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/3
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/4
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
```

```
interface GE1/0/5
description *** Printer ***
port link-type access
port default vlan 597
trust dscp
#
interface GE1/0/6
port link-type access
port default vlan 6
trust dscp
nd snooping max-user-number 0
#
interface GE1/0/7
port link-type access
port default vlan 6
trust dscp
nd snooping max-user-number 0
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/8
port link-type access
port default vlan 6
trust dscp
nd snooping max-user-number 0
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/9
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/10
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/11
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/12
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/13
description **Direct-to-SYScon-MUX**
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/14
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/15
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/16
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/17
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/18
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/19
```

```
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/20
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/21
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/22
port link-type hybrid
port hybrid tagged vlan 2 to 4094
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/23
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/24
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/25
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
undo poe enable
#
interface GE1/0/26
port link-type hybrid
```

```
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/27
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/28
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/29
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/30
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/31
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/32
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/33
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
```

```
#
interface GE1/0/34
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/35
port link-type hybrid
port hybrid tagged vlan 2 to 54 56 to 499 601
to 4094
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/36
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/37
port link-type hybrid
port hybrid tagged vlan 2 to 54 56 to 499 601
to 4094
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/38
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/39
port link-type hybrid
port hybrid tagged vlan 2 to 54 56 to 499 601
to 4094
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/40
port link-type hybrid

trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/41
port link-type hybrid
port hybrid tagged vlan 2 to 54 56 to 499 601
to 4094
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/42
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/43
description *** Huawei Cloud AP ***
port link-type trunk
port trunk pvid vlan 1100
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 55 500 to 600
trust dscp
#
interface GE1/0/44
description *** Huawei Cloud AP ***
port link-type trunk
port trunk pvid vlan 1100
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 55 500 to 600
trust dscp
#
interface GE1/0/45
description *** Huawei Cloud AP ***
port link-type trunk
port trunk pvid vlan 1100
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 55 500 to 600
trust dscp
#
interface GE1/0/46
description *** Huawei Cloud AP ***
port link-type trunk
```

```
port trunk pvid vlan 1100
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 55 500 to 600
trust dscp
#
interface GE1/0/47
port link-type access
port default vlan 6
trust dscp
nd snooping max-user-number 0
#
interface GE1/0/48
port link-type trunk
port trunk allow-pass vlan 2 to 4094
nd snooping max-user-number 0
#
interface GE1/0/49
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/50
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/51
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface GE1/0/52
port link-type hybrid
trust dscp
authentication-profile
Auth42f3b387e927016d4df7
#
interface NULL0
#
ip route-static 0.0.0.0 0.0.0.0 10.220.20.254
preference 1
ip route-static 10.0.0.0 255.0.0.0 10.135.30.1
```

```
ip route-static 10.212.135.0 255.255.255.0
10.0.0.1
ip route-static 172.16.0.0 255.255.0.0
10.135.30.1
#
snmp-agent local-engineid
800007DB03605375BF325F
#
snmp-agent sys-info version v3
#
undo snmp-agent protocol source-status all-
interface
undo snmp-agent protocol source-status ipv6
all-interface
#
undo snmp-agent proxy protocol source-
status all-interface
undo snmp-agent proxy protocol source-
status ipv6 all-interface
#
lldp auto-vlan 1100 sensor ap
lldp auto-vlan tagged 500 to 600 sensor ap
#
stelnet server enable
snetconf server enable
ssh server rsa-key min-length 3072
ssh user cebadmin
ssh user cebadmin authentication-type
password
ssh user cebadmin service-type stelnet
ssh user huawei
ssh user huawei authentication-type x509v3-
rsa
ssh user huawei assign pki default
ssh user huawei service-type snetconf
ssh server-source all-interface
ssh ipv6 server-source all-interface
ssh server assign pki default
ssh authorization-type default root
#
ssh server cipher aes256_gcm aes128_gcm
aes256_ctr aes192_ctr aes128_ctr
ssh server hmac sha2_512 sha2_256
```

```
ssh server key-exchange
dh_group_exchange_sha256
dh_group16_sha512 curve25519_sha256
#
ssh server publickey ecc x509v3-ssh-rsa
rsa_sha2_256 rsa_sha2_512
#
ssh server dh-exchange min-len 3072
#
ssh client first-time enable
#
ssh client publickey x509v3-ssh-rsa
rsa_sha2_256 rsa_sha2_512 x509v3-rsa2048-
sha256
#
ssh client cipher aes256_gcm aes128_gcm
aes256_ctr aes192_ctr aes128_ctr
ssh client hmac sha2_512 sha2_256
ssh client key-exchange
dh_group_exchange_sha256
dh_group16_sha512 curve25519_sha256
#
user-interface maximum-vty 5
#
user-interface con 0
authentication-mode aaa
idle-timeout 30 0
#
user-interface vty 0 4
authentication-mode aaa
user privilege level 3
protocol inbound ssh
#
netconf
callhome default-callhome
endpoint vlanif1100_10.220.20.31
peer-ip 10.128.20.112 port 10020
#
ztp domain-type registration-center domain
register.naas.huawei.com port 10020
#
grpc
#
grpc client
ssl-policy default
```

```
#
telemetry
#
sensor-group CloudManagerCollectPolicy0
sensor-path huawei-remote-unit-
mng:remote-unit-mng/remote-unit-
status/basic-data/device-datas/device-data
sensor-path huawei-remote-unit-
mng:remote-unit-mng/remote-unit-
status/basic-data/interfaces/interface
sensor-path huawei-remote-unit-
mng:remote-unit-mng/remote-unit-
status/basic-data/optical-modules/optical-
module
#
sensor-group CloudManagerCollectPolicy300
sensor-path huawei-aaa:aaa/access-user-
qrys/access-user-qry
sensor-path huawei-aaa:aaa/access-user-
records/access-user-record
sensor-path huawei-arp-security:arp-
security/arp-snooping-records/arp-snooping-
record
sensor-path huawei-cpu-memory:cpu-
memory/board-cpu-infos/board-cpu-info
sensor-path huawei-cpu-memory:cpu-
memory/board-memory-infos/board-
memory-info
sensor-path huawei-cpu-memory:cpu-
memory/board-storage-partition-infos/board-
storage-partition-info
sensor-path huawei-devm-poe:devm-
poe/poes/poe/ports/port
sensor-path huawei-
ifm:ifm/interfaces/interface/common-
statistics
sensor-path huawei-
ifm:ifm/interfaces/interface/mib-
statistics/huawei-pic:eth-port-err-sts
sensor-path huawei-remote-unit-
mng:remote-unit-mng/remote-unit-
status/detail-data/device-datas/device-data
sensor-path huawei-remote-unit-
mng:remote-unit-mng/remote-unit-
status/detail-data/interfaces/interface
```

```
sensor-path huawei-remote-unit-
mng:remote-unit-mng/remote-unit-
status/detail-data/optical-modules/optical-
module
sensor-path huawei-
vlan:vlan/vlans/vlan/vlan-statistic
depth 2
#
destination-group
CloudManagerCollectConnection
ipv4-address 10.128.20.112 port 10034
#
subscription AC0000
protocol grpc
pack enable
sensor-group CloudManagerCollectPolicy0
sample-interval 0
destination-group
CloudManagerCollectConnection
#
subscription AC0300
protocol grpc
pack enable
sensor-group CloudManagerCollectPolicy300
sample-interval 300000
destination-group
CloudManagerCollectConnection
#
pnp startup-vlan send enable
#
web-manager enable port 8443
web-manager http forward enable
#
return
```

9.3 iMaster Configuration (Sample)

Admission / Admission Policy / Authentication and Authorization

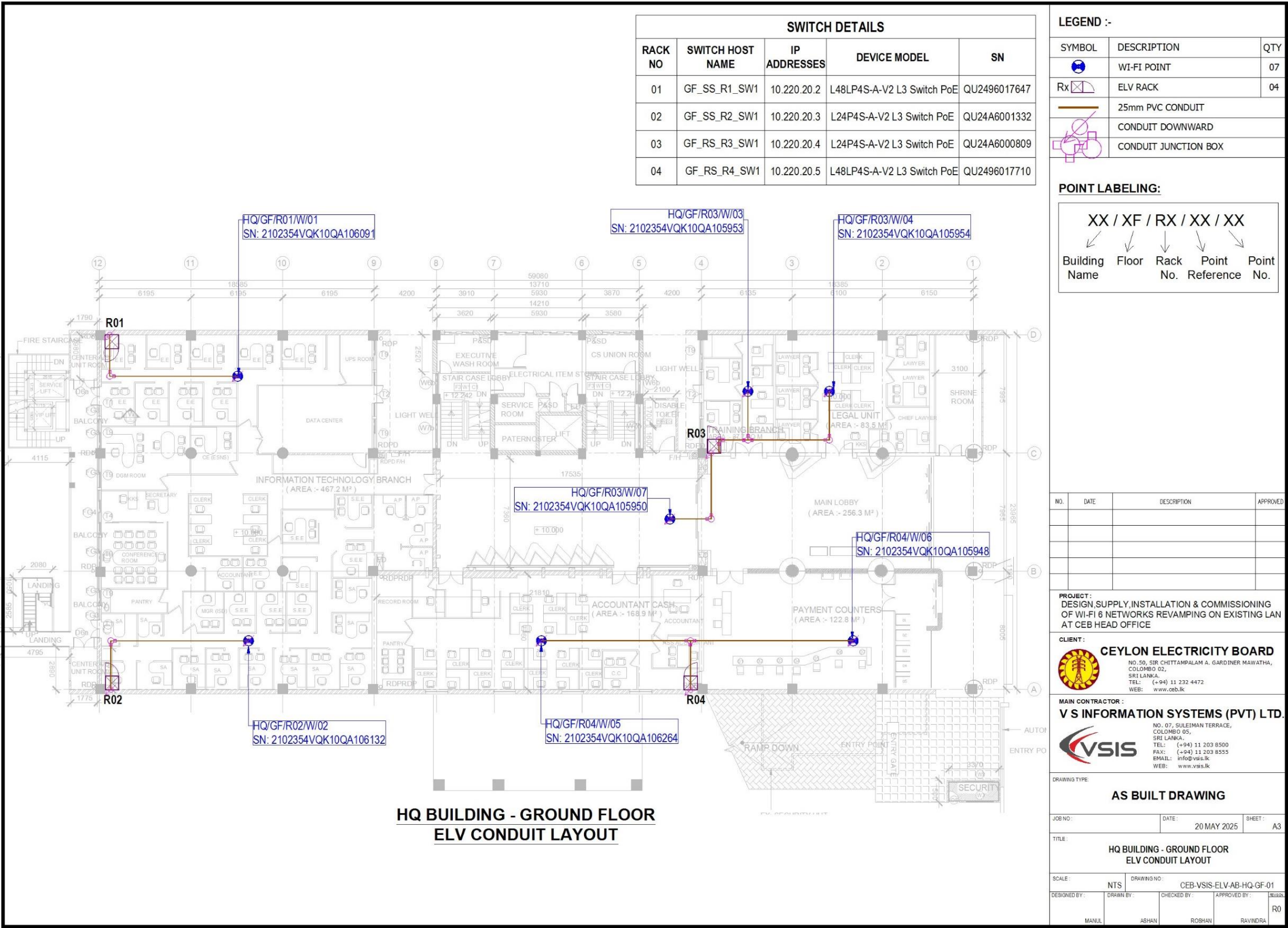
[Authentication Rule](#) | [Authorization Result](#) | [Authorization Rule](#) | [Policy Element](#) | [Multi-level RADIUS Relay](#) | [Network Access Policy](#)

☐	35	Wireless - CE (Tx Li...	User access authen...	Wireless	User group ROOTICEB-Users\CEB_Users\AGM (Tx - N WO\DSM (Trans. & Generation Planning)\Tr SSID CEB_Employees	Assign-vlan-514	156	Enabled					
Expand all													
☐	36	Wired - DGM(Trans...	User access authen...	Wired	User group ROOTICEB-Users\CEB_Users\AGM (Tx - N WO\DSM (Trans. & Generation Planning)\Tr	Assign-vlan-516	41	Enabled					
Expand all													
☐	37	Wireless - DGM(Tra...	User access authen...	Wireless	User group ROOTICEB-Users\CEB_Users\AGM (Tx - N WO\DSM (Trans. & Generation Planning)\Tr SSID CEB_Employees	Assign-vlan-516	848	Enabled					
Expand all													
☐	38	Wireless-DGM(Pers...	User access authen...	Wireless	User group ROOTICEB-Users\CEB_Users\AGM (Perso nnel) SSID CEB_Employees	Assign-vlan-530	377	Enabled					
Expand all													
☐	39	Wireless - CE(Tx-Pl...	User access authen...	Wireless	User group ROOTICEB-Users\CEB_Users\AGM (Tx - N WO\DSM (Trans. & Generation Planning)\Tr SSID CEB_Employees	Assign-vlan-517	0	Enabled					
Expand all													
☐	40	Wireless - CE (Gen...	User access authen...	Wireless	User group ROOTICEB-Users\CEB_Users\AGM (Tx - N WO\DSM (Trans. & Generation Planning)\Tr SSID CEB_Employees	Assign-vlan-518	0	Enabled					
Expand all													
☐	41	Wireless - CE (Gen...	User access authen...	Wireless	User group ROOTICEB-Users\CEB_Users\AGM (Tx - N WO\DSM (Trans. & Generation Planning)\Tr SSID CEB_Employees	Assign-vlan-519	0	Enabled					
Expand all													

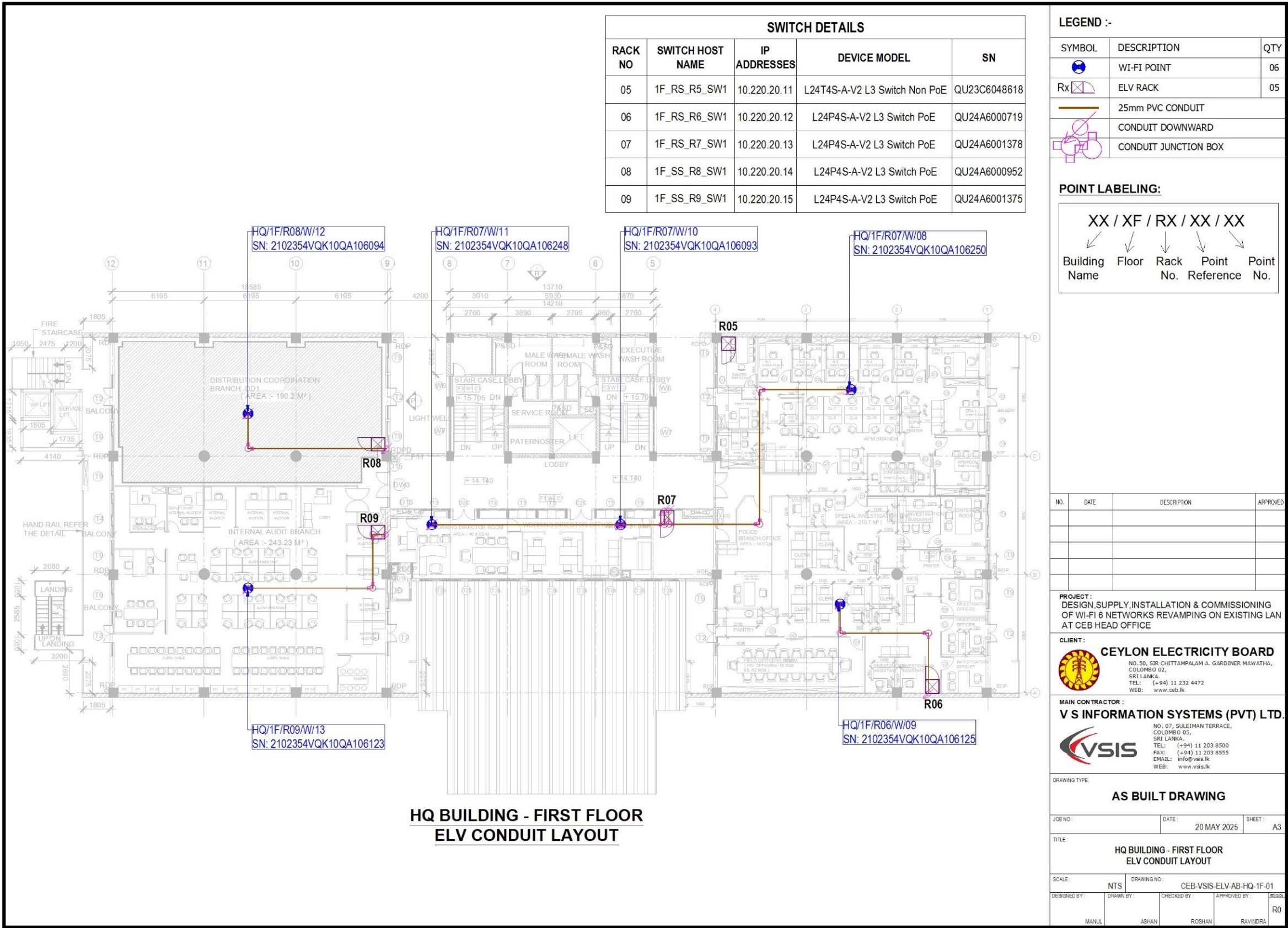
10 Drawings

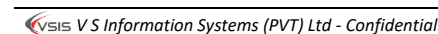
Intentionally left blank.....

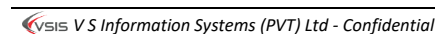
10.1.1 Head Quarters Building –Groud Floor

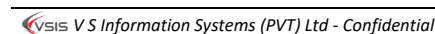


10.1.2 Head Quarters Building –First Floor

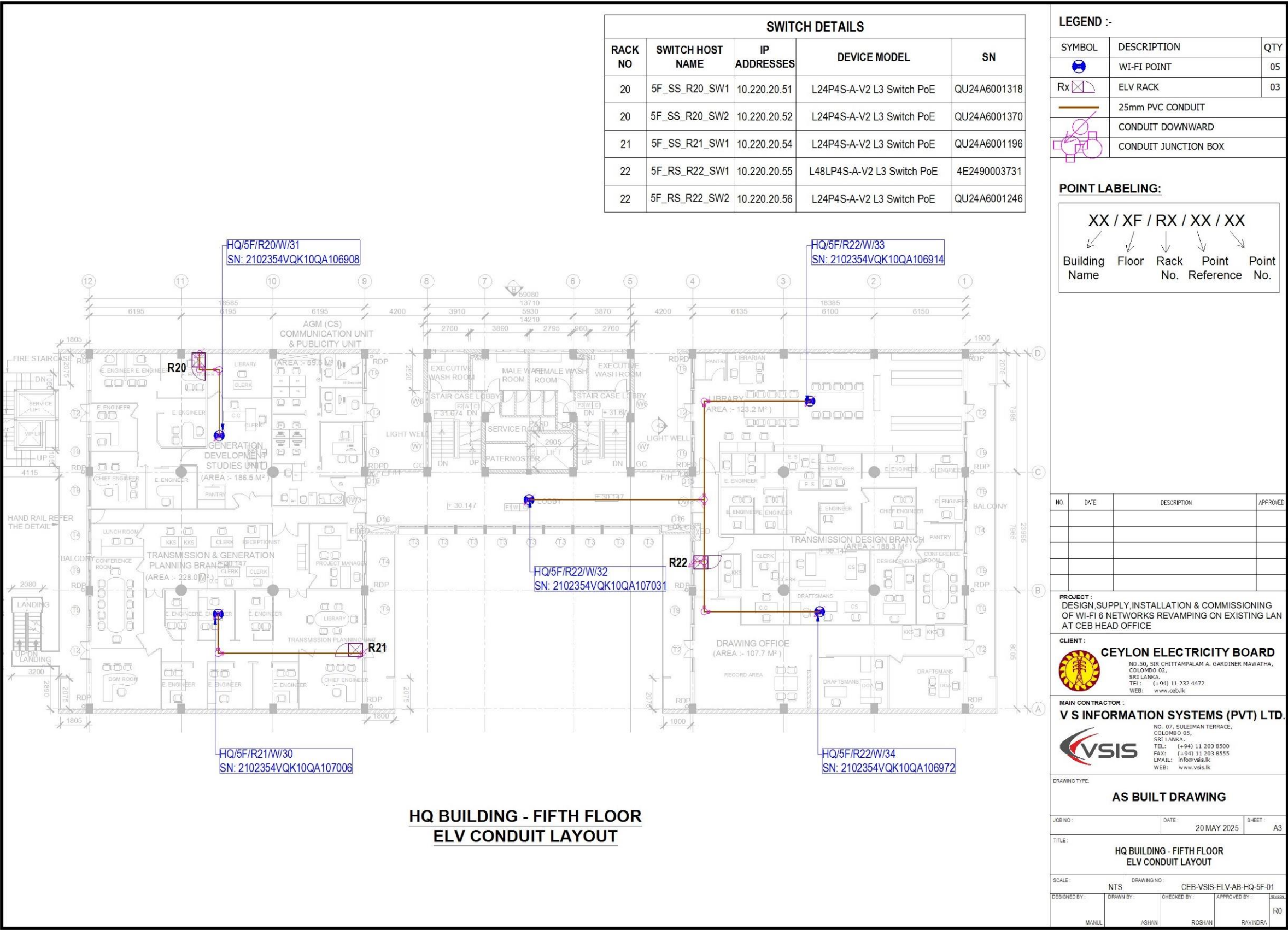




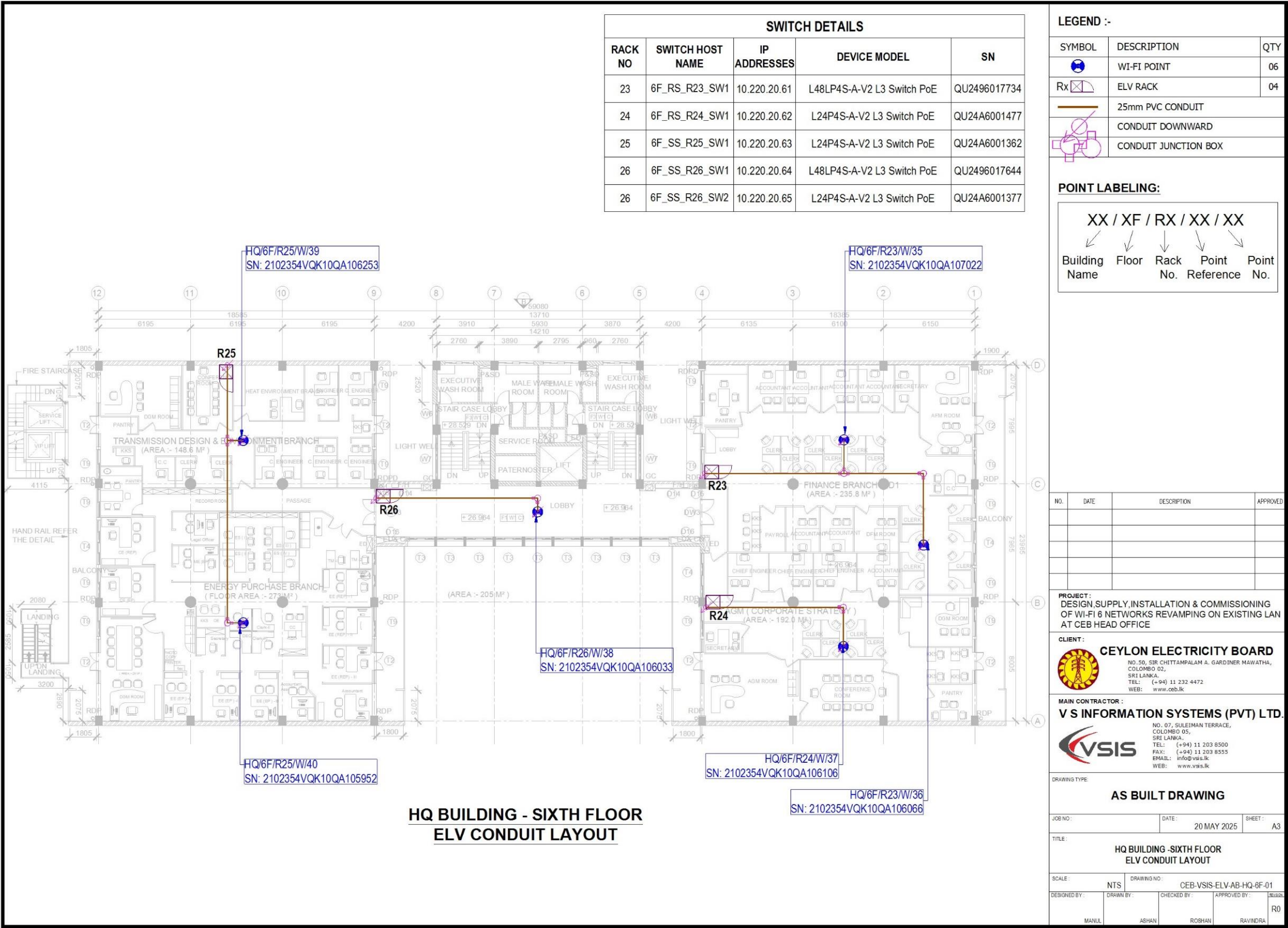




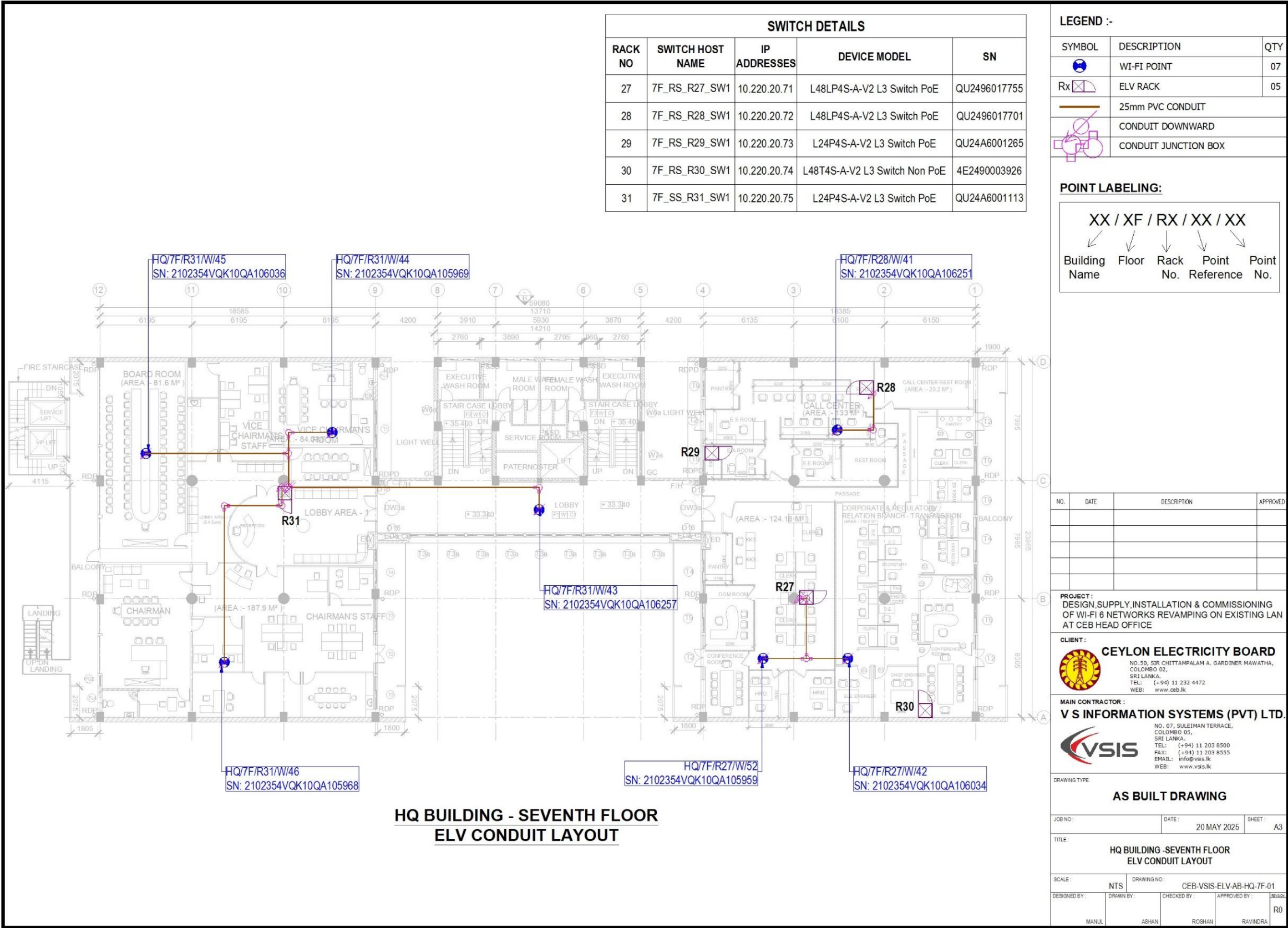
10.1.6 Head Quarters Building –Fifth Floor



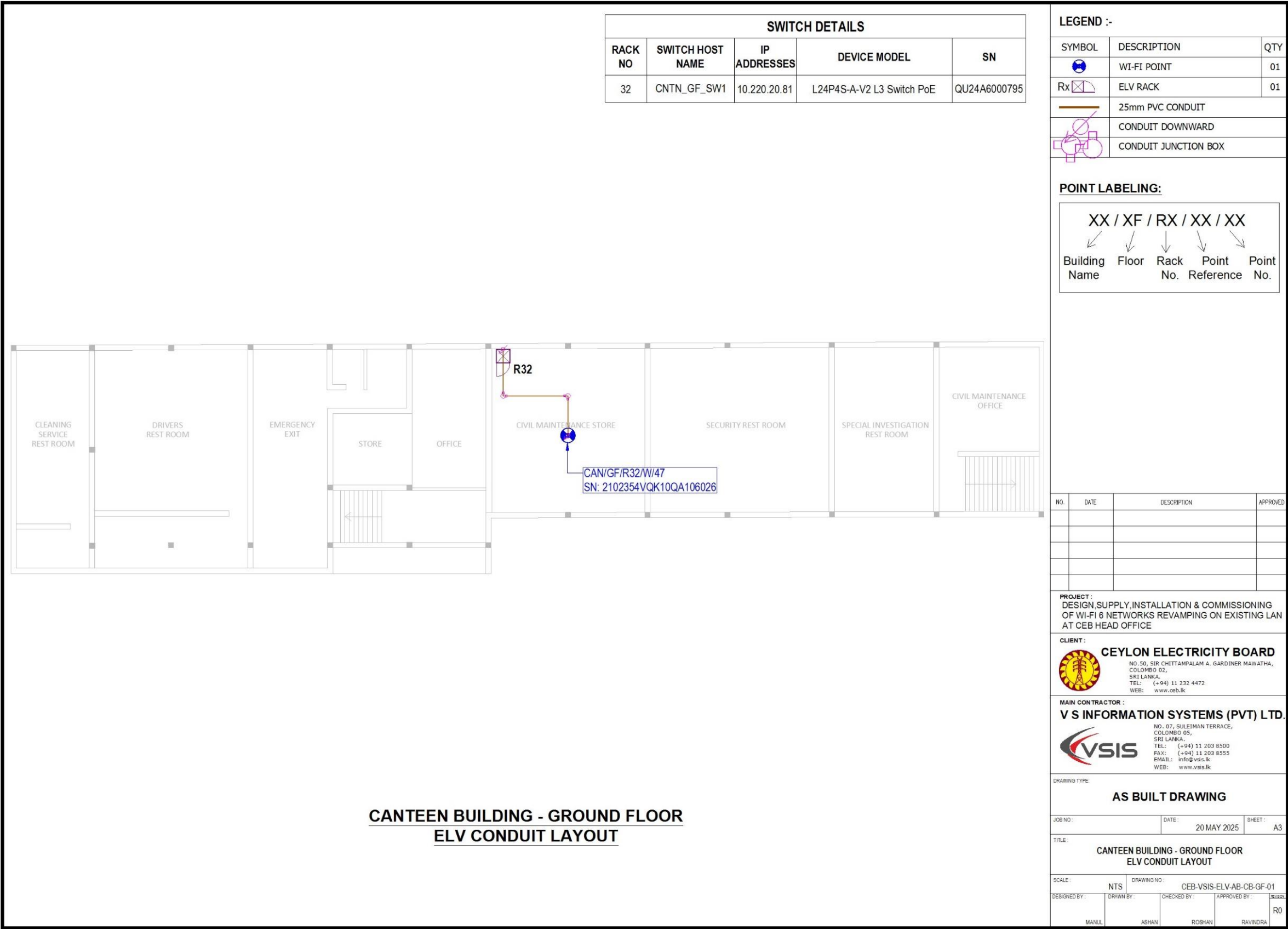
10.1.7 Head Quarters Building –Sith Floor



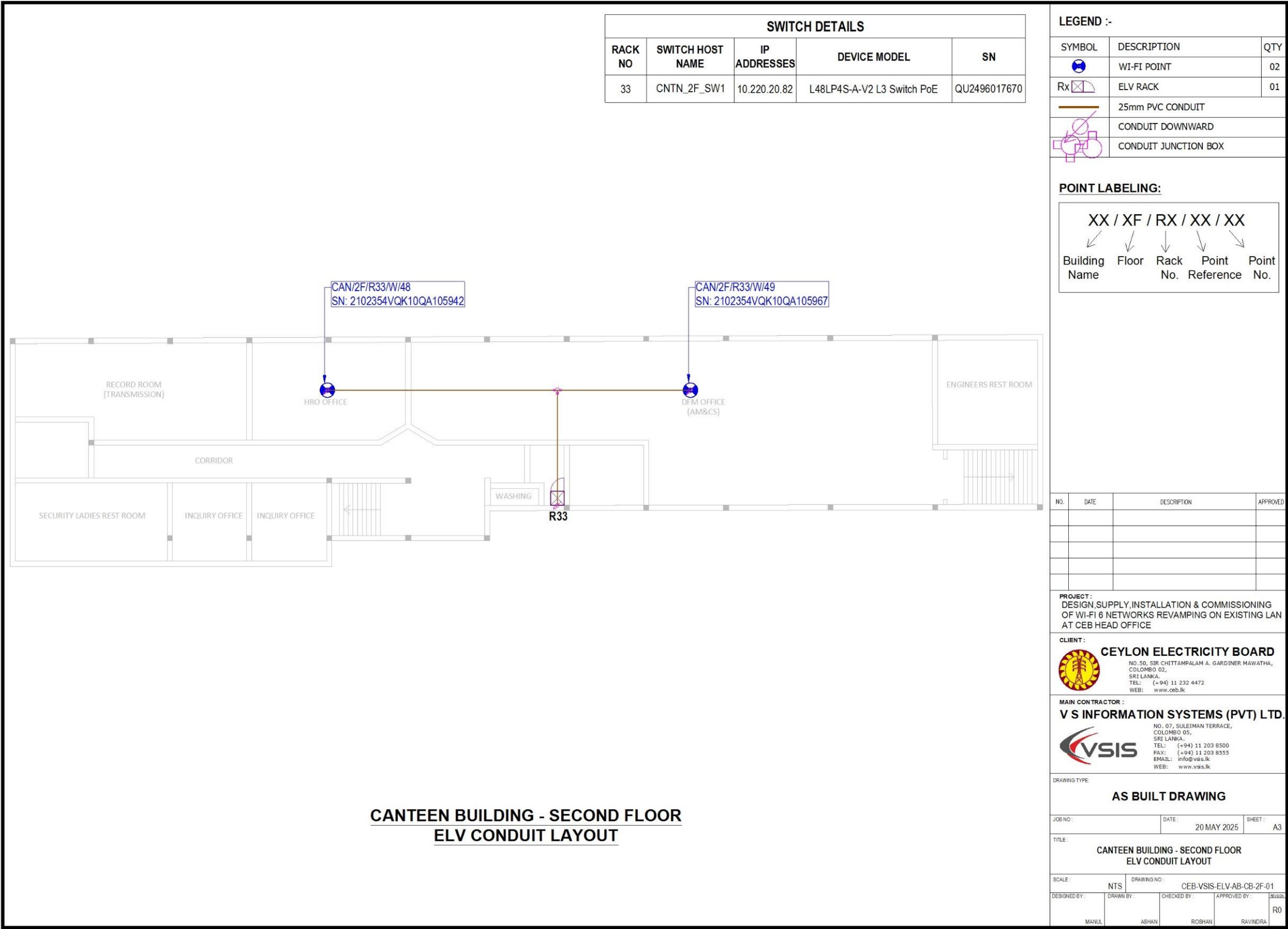
10.1.8 Head Quarters Building –Seventh Floor



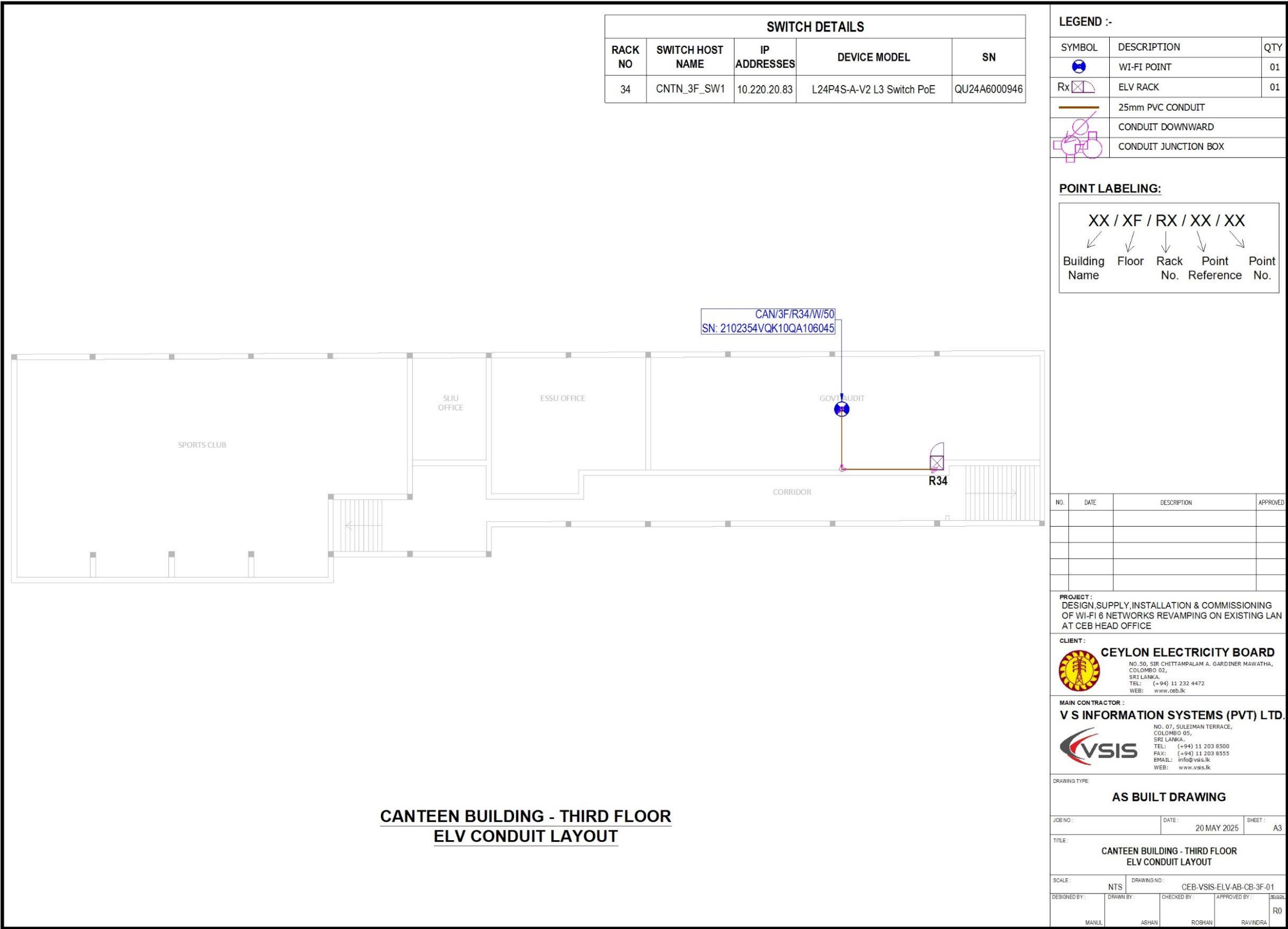
10.1.9 Canteen Building –Ground Floor



10.1.10 Canteen Building –Second Floor

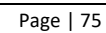


10.1.11 Canteen Building –Third Floor

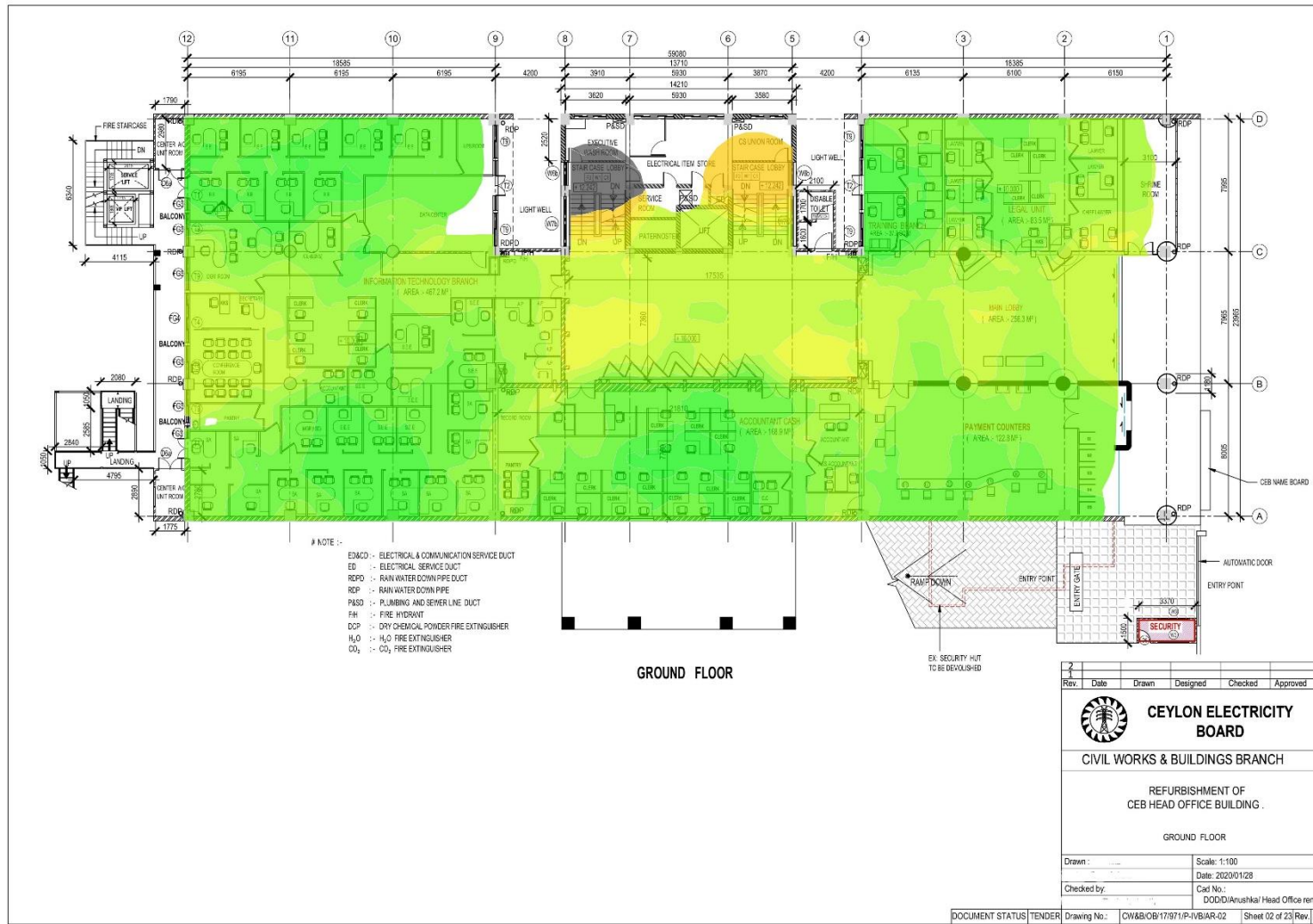


10.2 Live Heat Maps

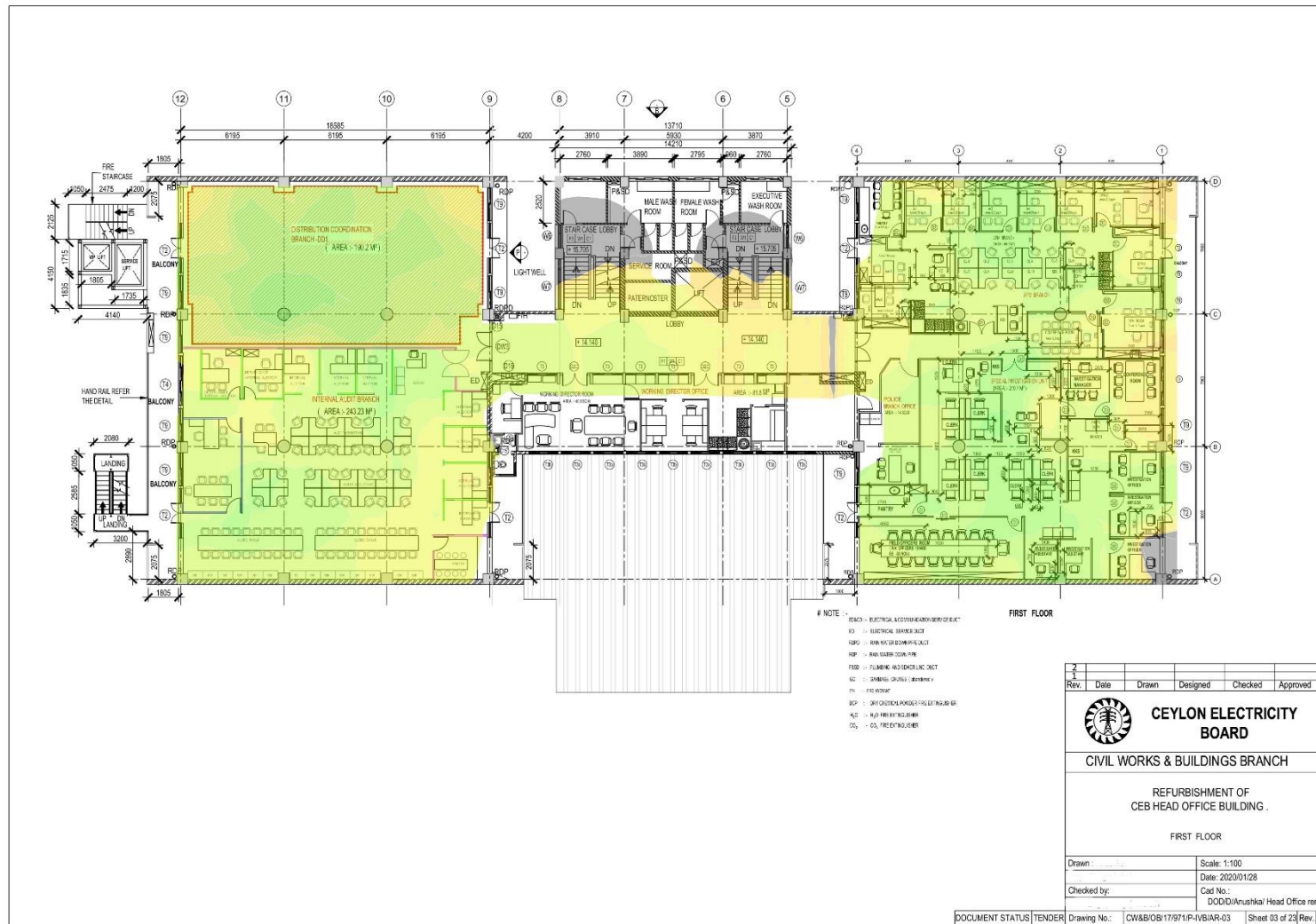
Intentionally left blank.....

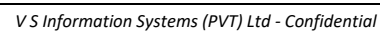


10.2.2 Heat Map - Head Quarters Building – Ground Floor 5 Gz Band

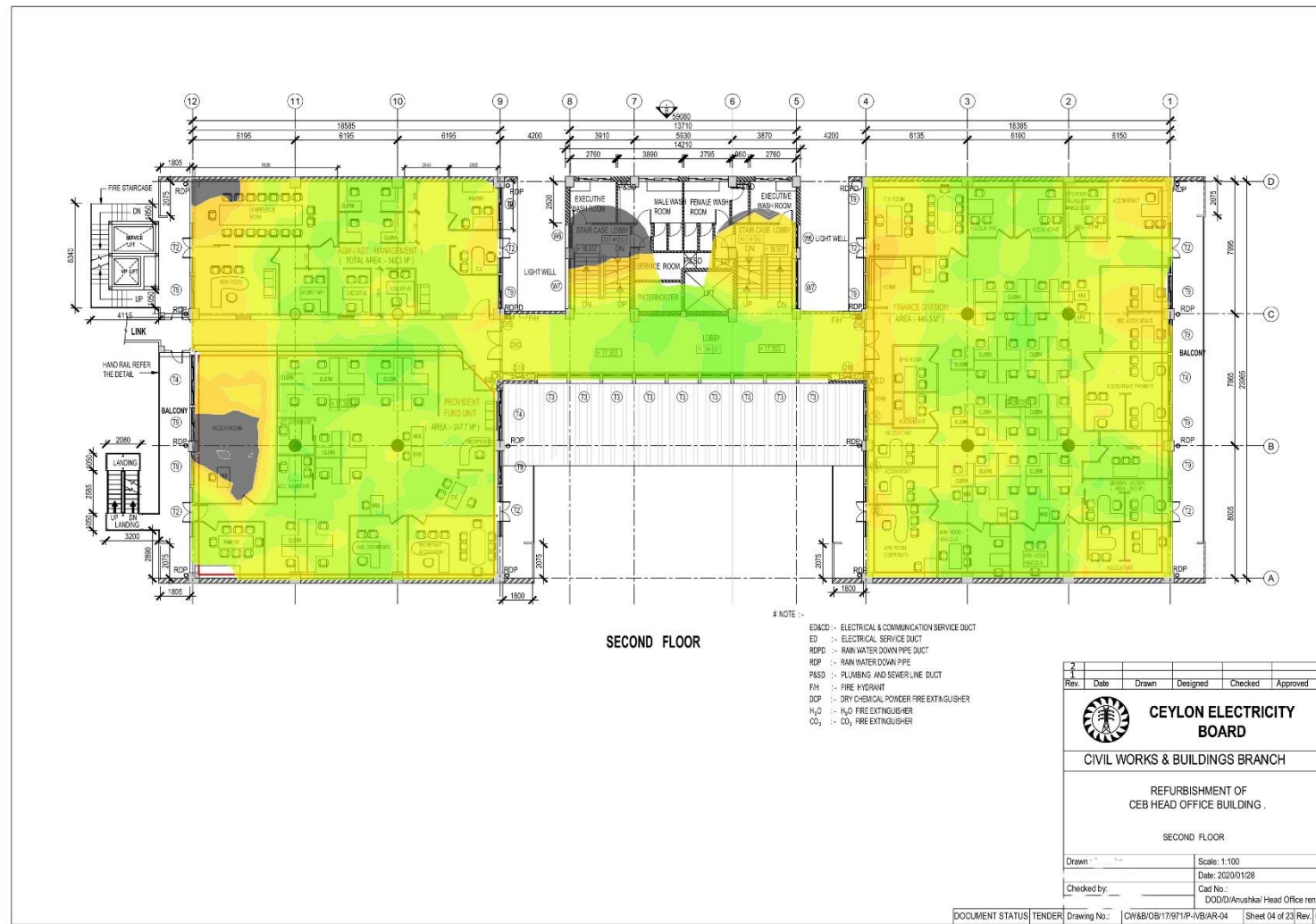


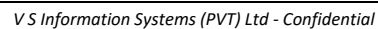
10.2.3 Heat Map – Head Quarters Building – First Floor – 2.4 GHz Band



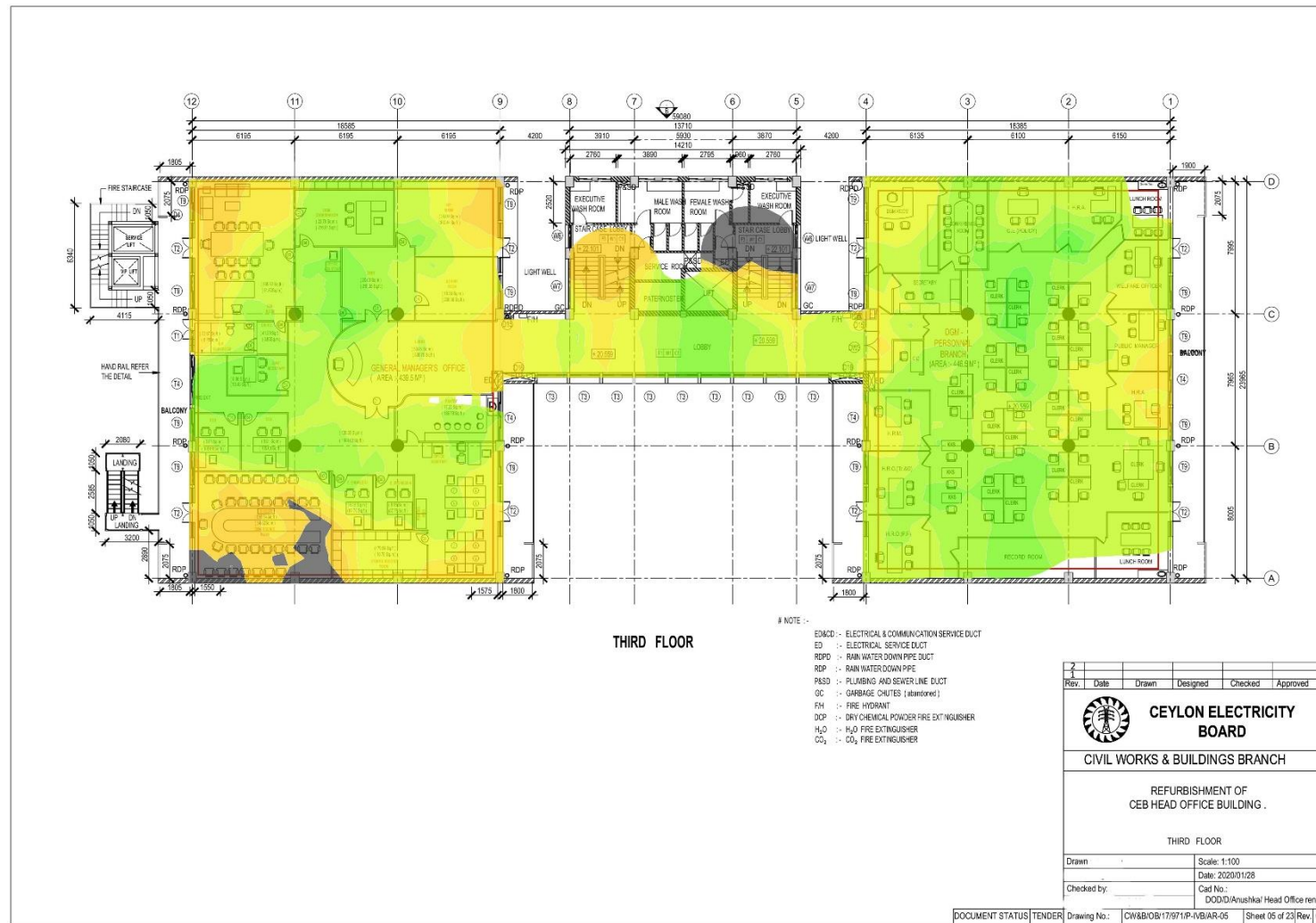


10.2.5 Heat Map – Head Quarters Building – Second Floor – 2.4 GHz Band

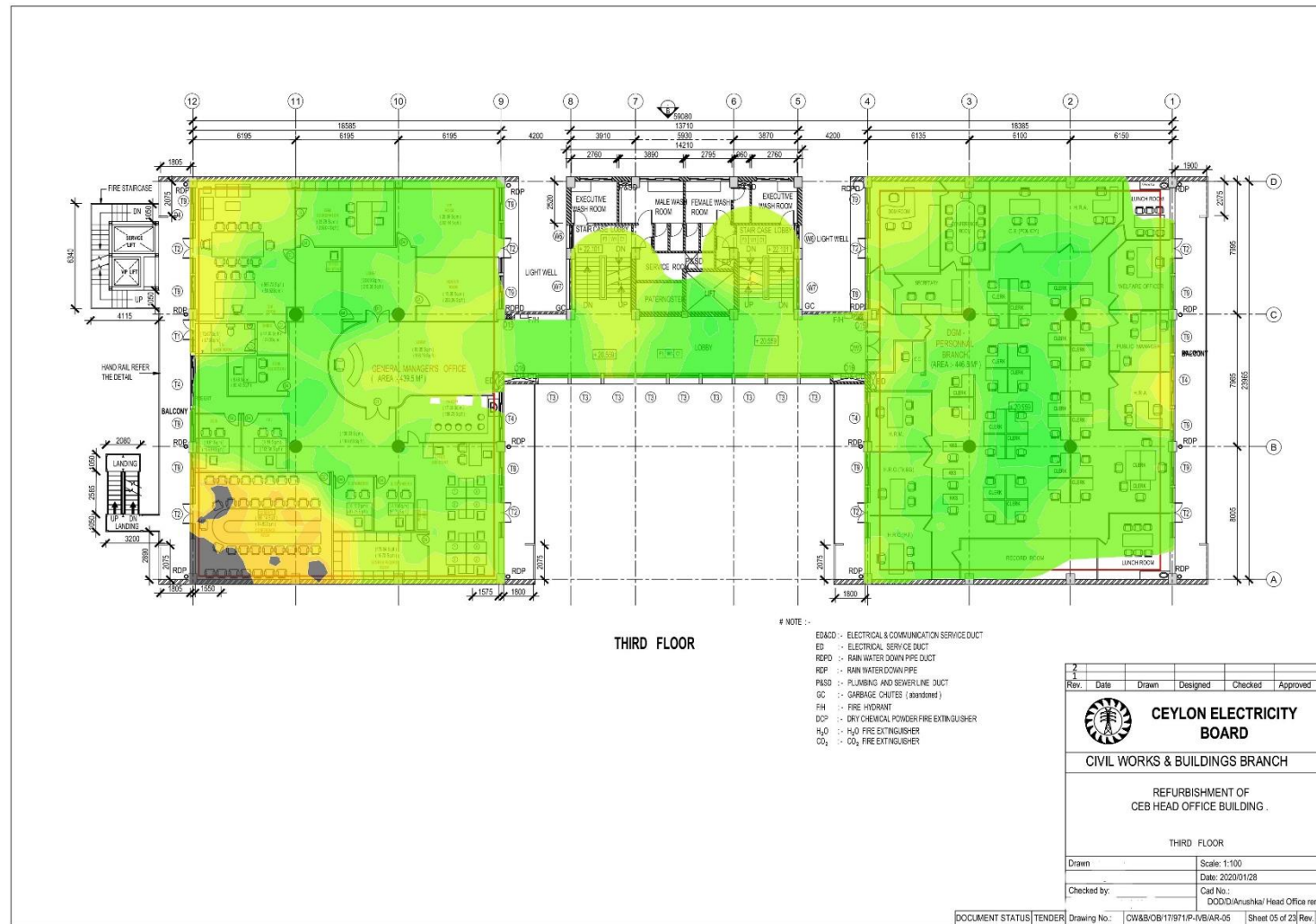




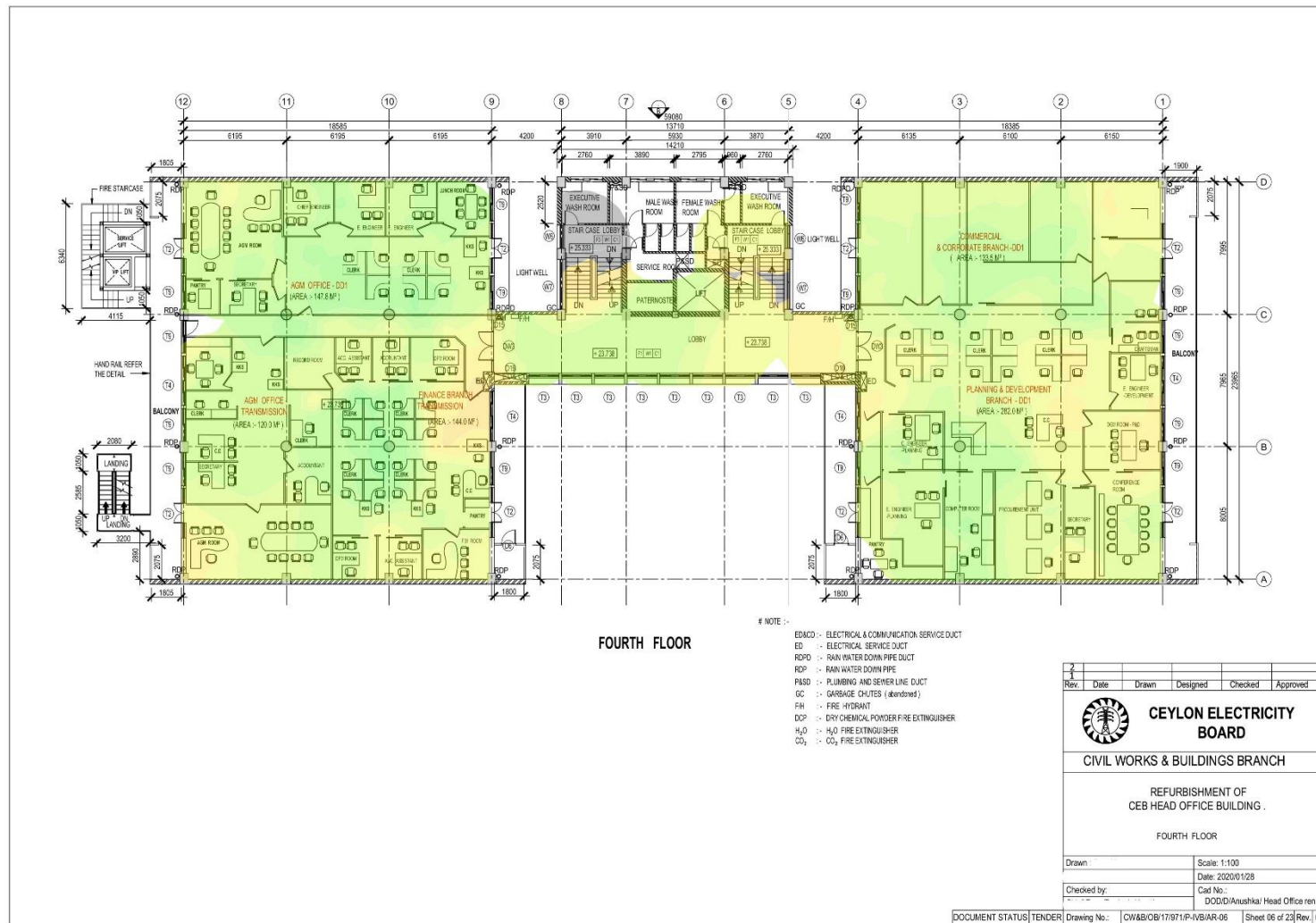
10.2.7 Heat Map – Head Quarters Building – Third Floor – 2.4 GHz Band



10.2.8 Heat Map – Head Quarters Building – Third Floor – 5 GHz Band



10.2.9 Heat Map – Head Quarters Building – Fourth Floor – 2.4 GHz Band



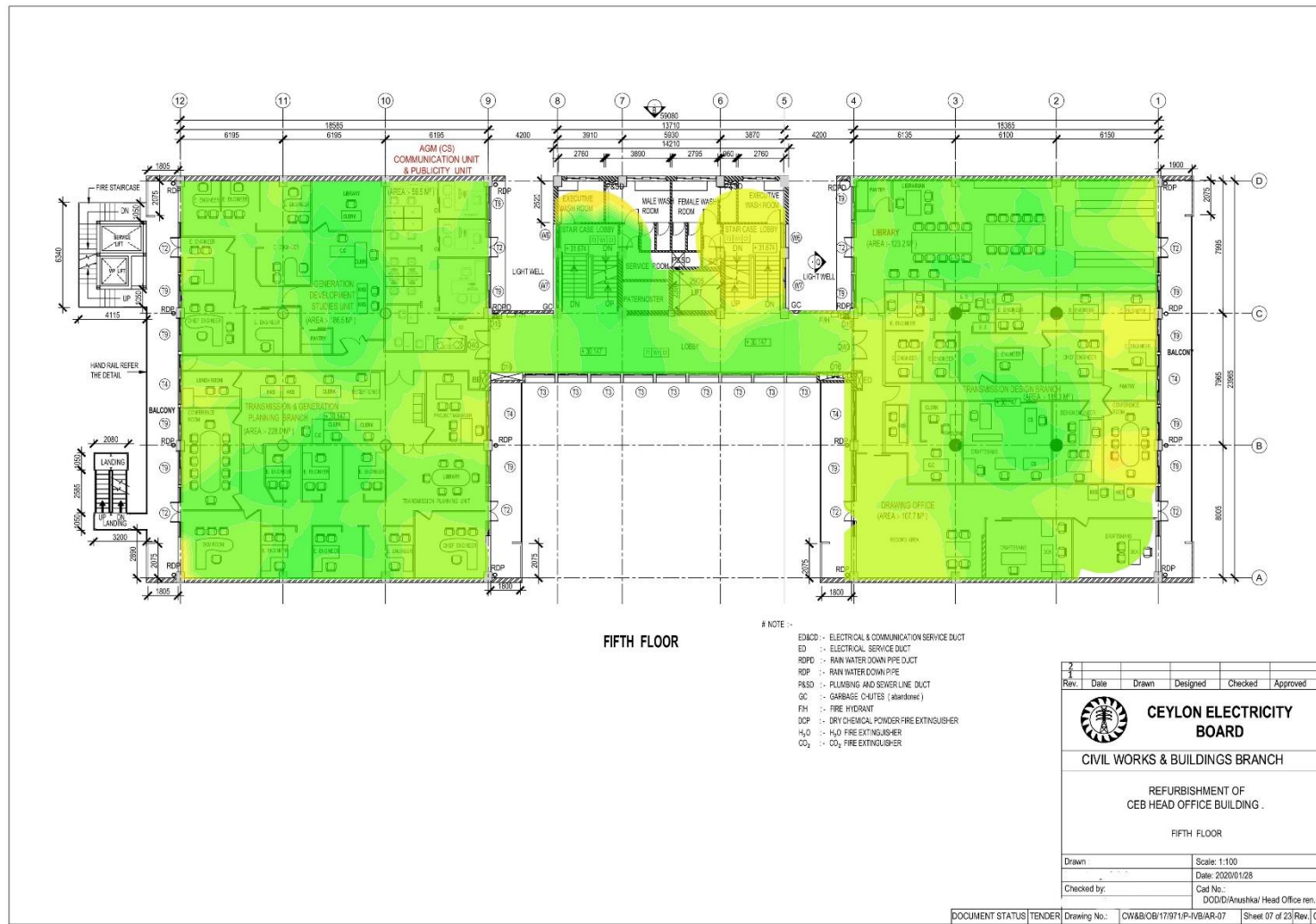
10.2.10 Heat Map – Head Quarters Building – Fourth Floor – 5 GHz Band



10.2.11 Heat Map – Head Quarters Building – Fifth Floor – 2.4 GHz Band



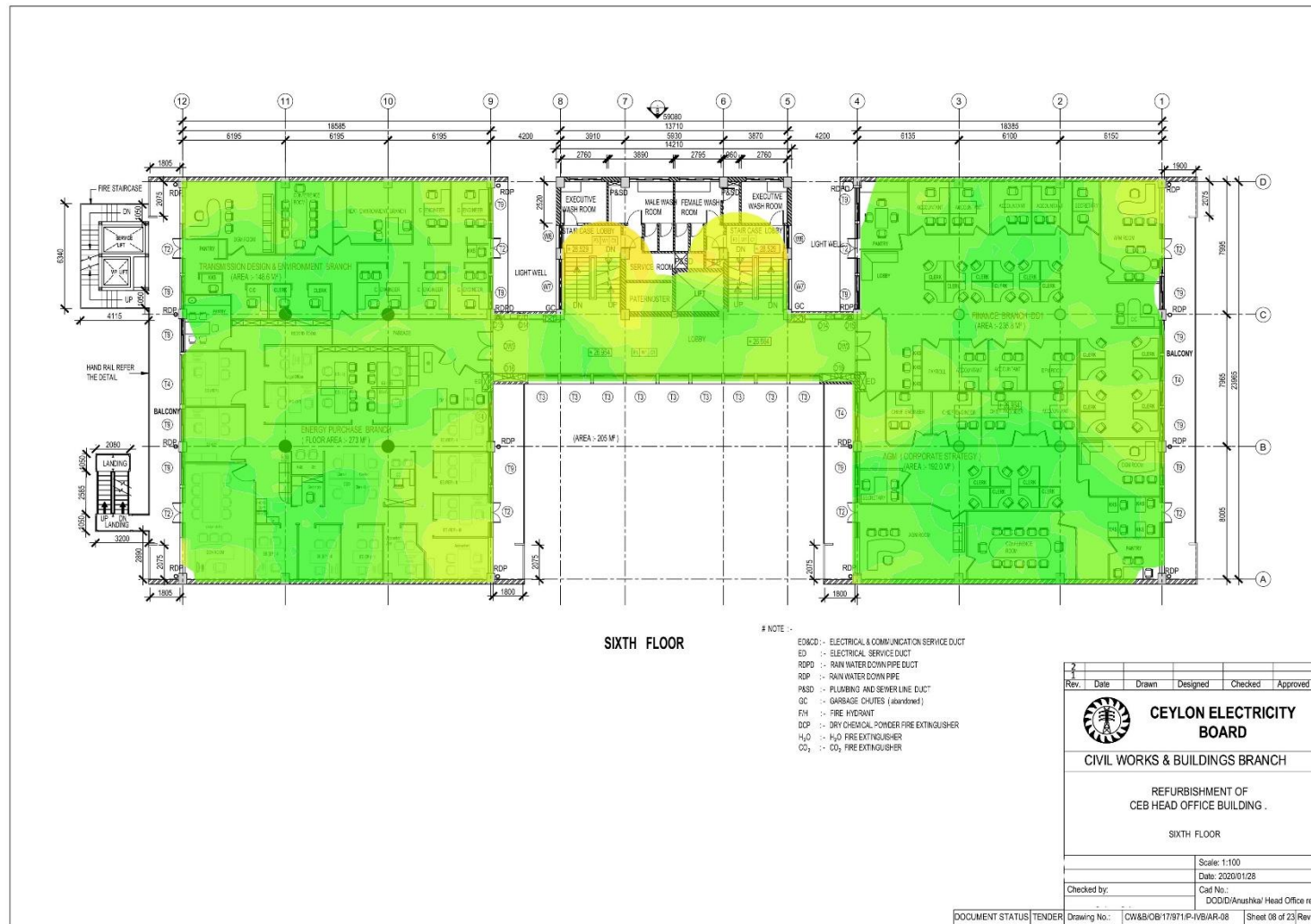
10.2.12 Heat Map – Head Quarters Building – Fifth Floor – 5 GHz Band



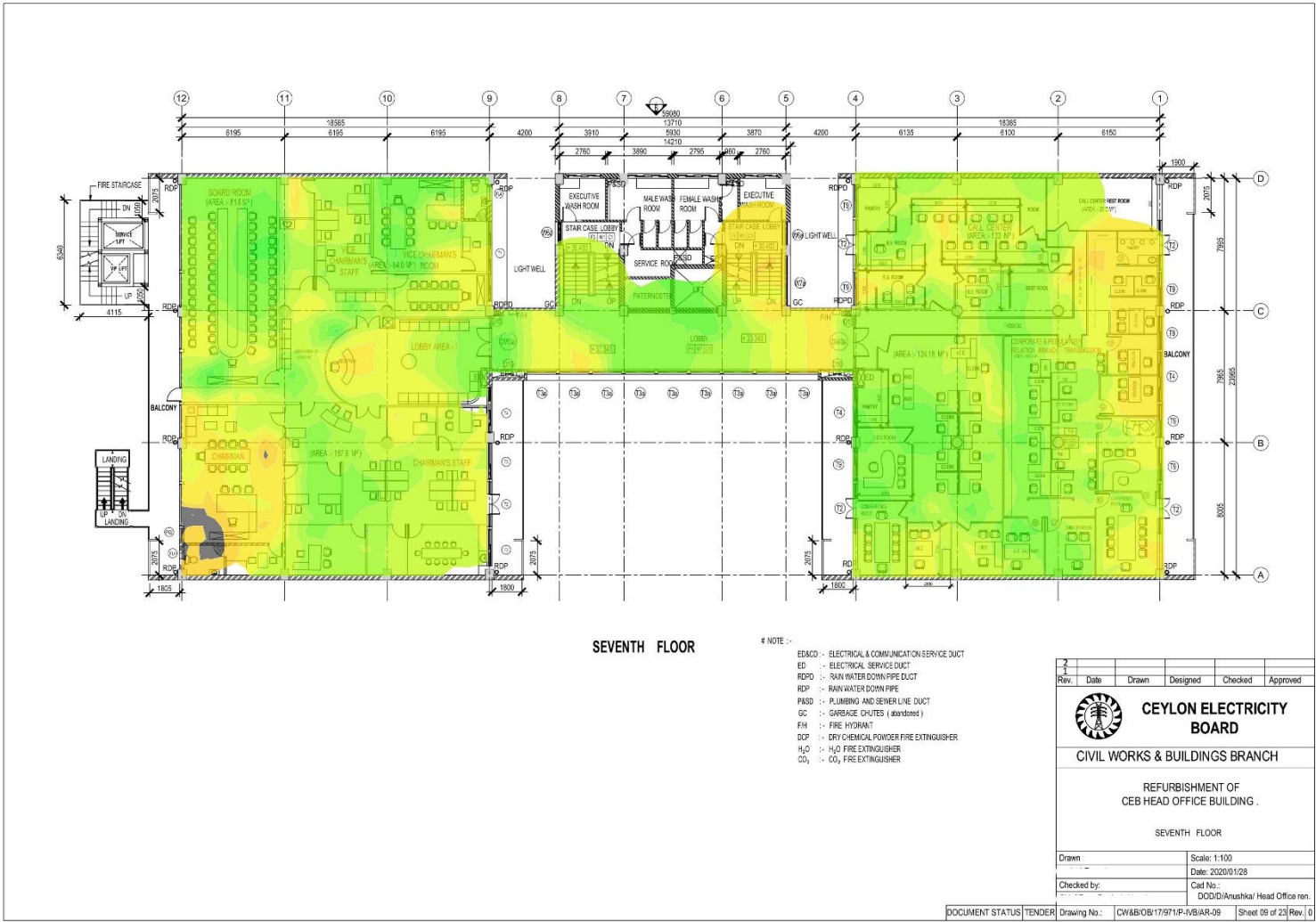
10.2.13 Heat Map – Head Quarters Building – Sixth Floor – 2.4 GHz Band



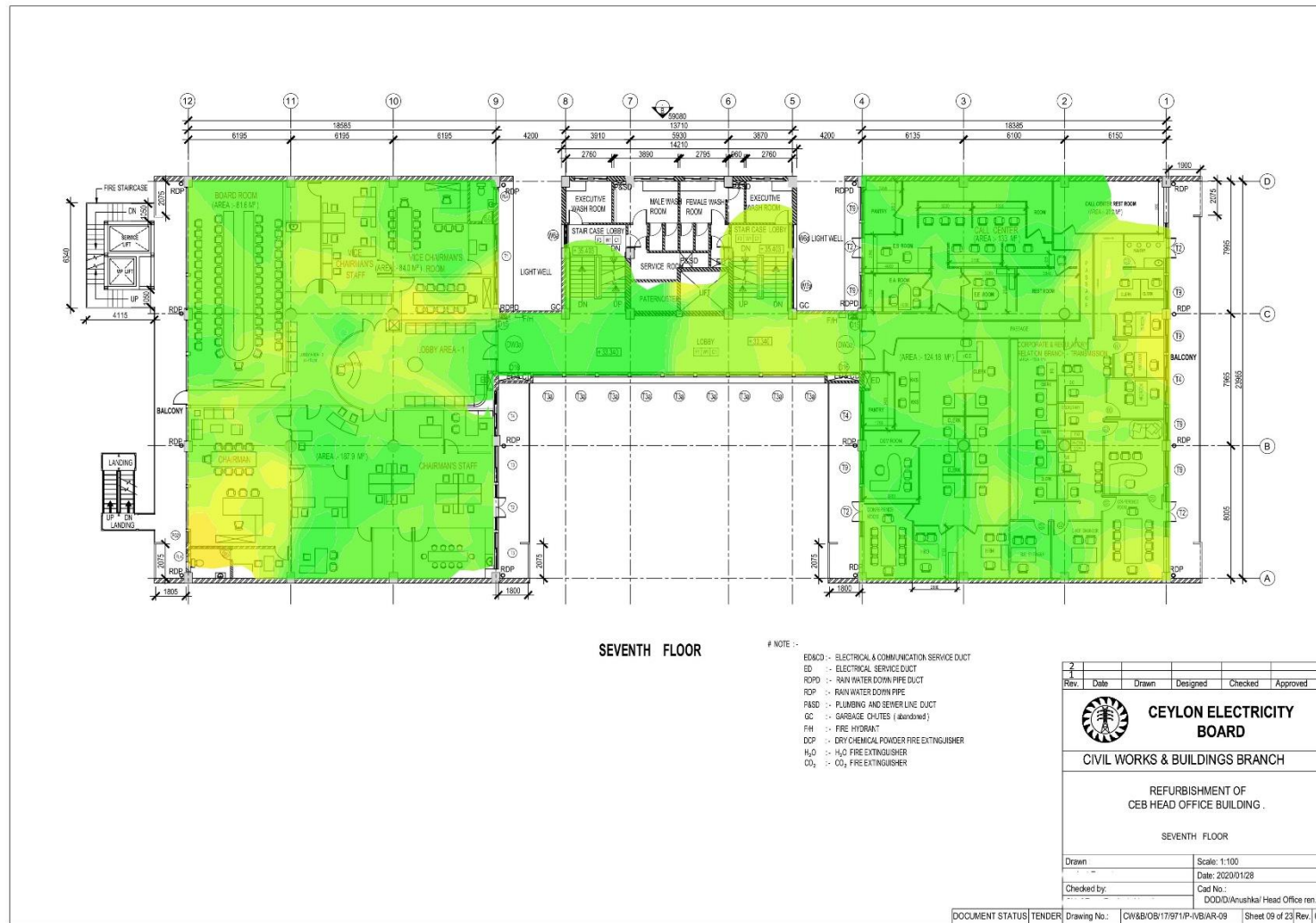
10.2.14 Heat Map – Head Quarters Building – Sixth Floor – 5 GHz Band



10.2.15 Heat Map – Head Quarters Building – Seventh Floor – 2.4 GHz Band



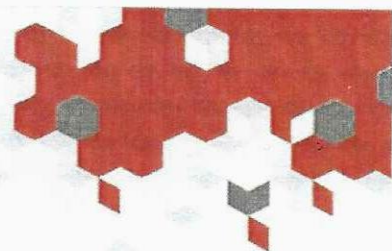
10.2.16 Heat Map – Head Quarters Building – Seventh Floor – 5 GHz Band



11 Administration Information

All administrative credentials and access details related to the network devices and management platforms have been compiled into a secured document and formally handed over to the authorized personnel of CEB, ensuring confidentiality and proper access control. This includes login credentials for the network switches and the Huawei iMaster NCE-Campus centralized management system.

12 Warranty Certifications



July 23rd, 2025

Ceylon Electricity Board Head Office,
No 50,
Sir C. A. Gardiner Mawatha,
Colombo 02.

Dear Sir,

SUPPLY, INSTALLATION AND COMMISSIONING OF WIFI NETWORK AND REVAMPING OF EXISTING LAN AT CEB HEAD OFFICE

It's a pleasure to inform you that V S Information Systems (Pvt) Ltd (VSIS) have completed the above said project successfully.

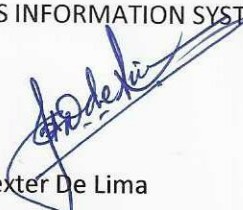
We vsis certify the warranty that the supplied equipment is covered by manufacturing defects up to a period of 3 Year commencing from **18th of July 2025**.

Warranty is applicable for manufacture's defects only. And it will not cover the damages made by physically, lighting/surge, act of god/ natural disasters or not kept in the recommended air conditioned environment.

Thanking and assuring you of our bet services at all times,

Yours faithfully,

V S INFORMATION SYSTEMS (PVT) LTD


Dexter De Lima
Consultant



V S Information Systems (Pvt) Ltd

No. 07, Sulaiman Terrace, Colombo 05, Sri Lanka

Telephone: +94 (11) 203 8500, +94 (11) 703 8555 Fax: +94 (11) 255 5805 Website: www.vsis.lk Reg. No. PY 65809

12.2 Warranty Information

Component	Premium
Warranty Period	Five (03) Years from the date of handed over (for all components)
Number of cases	Unlimited
Service Delivery	a) Response 24x7 telephone and on-site support. (Provide 24x7 available helpdesk phone number) b) Response time - Two (2) hour onsite response time c) Technical Review - By-Annually

12.3 Contact Details

Name	Designation & Responsibility	Contact Numbers
Mr. Sandaruwan Basnayake	Electrical Engineer– Ceylon Electricity Board	Mob. +94 (71) 238 6177
Mr. Nieranjan Rathnayake	Electrical Engineer – Ceylon Electricity Board	Mob. +94 (71) 018 3640
Mr. Dexter De Lima	Consultant(VSIS)	Gen. +94 11 2038 500 Mob. +94 (77) 7620 227
Mr. Krishal Kahalage	Senior Network Systems Engineer(VSIS)	Gen. +94 11 2038 500 Mob. +94 (75) 7045 780
Mr. Manul De Silva	Network Systems Engineer(VSIS)	Gen. +94 11 2038 500 Mob. +94 (77) 8853 572

12.4 Support Contact Information

Access to Service Desk	8 am – 5 pm on Business Days	After Office-Hours and Non-Business Days (24x7)
Telephone	+94-11-203-8555	+94-11-203-8555
Onsite Engineer	Contact will be given once the ticket is assigned	
E-mail	support@vsis.lk	
Fax	+94-11-255-5805	
Web	www.vsis.lk	

12.5 Support Escalation Information

Name	Designation & Responsibility	Contact Numbers
Service Desk Agent	Service Desk Agent	Gen. +94-11-203-8555
Mr. Krishal Kahalage	Senior Network Systems Engineer(VSIS)	Gen. +94 11 2038 500 Mob. +94 (75) 7045 780
Mr. Manul De Silva	Network Systems Engineer(VSIS)	Gen. +94 11 2038 500 Mob. +94 (77) 8853 572
Mr. Dexter De Lima	Consultant(VSIS)	Gen. +94 11 2038 500 Mob. +94 (77) 7620 227
Mr. Chandika Nettasinghe	Director(VSIS)	Gen. +94 11 2038 500 Mob. +94 (77) 7333 176

12.6 Project Team

Implementation Team	
Krishal Kahagalage	Senior Network Systems Engineer
Roshan Wadimuna	Senior Network Systems Engineer
Manul De Silva	Network Systems Engineer
Bhanuka Alawaththa	Associate Network Systems Engineer
Hamzath Asan	Network Associate
Dinuwan Randunu	Team Lead - Cybersecurity
Gayathra Hashan	Network Engineering Trainee
Minindu Thiranjaya	Network Engineering Trainee
Ravindra Pathirana	Head of Delivery Management –Enterprise System Solutions
Project Management Team	
Prashadi Jayasinghe	Project Manager
Mayuran Muhas	Head of Project Management

13 Certificate of Acceptance

Intentionally left blank.....

CERTIFICATE OF ACCEPTANCE

Project Title: DESIGN, SUPPLY, INSTALLATION AND COMMISSIONING OF WIFI NETWORK AND REVAMPING OF EXISTING LAN AT CEB HEAD OFFICE

Sales Order NO: 2024-Sep-0918

Location: No 50, C.A Gardiner Mawatha, Colombo 02, Sri Lanka.

Date: 23rd July 2025

Customer/Owner: Ceylon Electricity Board Head Office

Contractor: V S Information Systems (Pvt) Ltd.
#7, Sulaiman Terrace, Colombo 00500, Sri Lanka.

Project Manager:

The Work performed under this Contract has been reviewed and found to be complete and has reached User Acceptance. The Date of User Acceptance of the Project is defined as the Date Certified by the Project Manager upon which the Project is fully completed in all accepts the VSIS's project work as completed to the fullest satisfaction. The Date of User Acceptance of the Project, or portion there of designated above, is also the basis for the commencement of the duration of applicable warranties required by the Contract Documents. This date shall correspond to the date of the Project Manager's approval on the final pay application unless otherwise agreed upon in writing.


Mr. Krishal Kahagalage
Project Engineer


Mr. Manul De Silva
Network Systems Engineer


Prashadi Jayasinghe
Project Manager



Authorized Signature:

Name:

Designation:

Date:

Customer:


Sandaruwan Basnayake
ER (Network & Security)

23rd July 2025

Ceylon Electricity Board (H.O.)

For DEPUTY GENERAL MANAGER
(INFORMATION TECHNOLOGY)
CORPORATE STRATEGY DIVISION
CEYLON ELECTRICITY BOARD

14 User Guides and Administrative Guides

1 AirEngine 5761-21 Access Point

<https://support.huawei.com/enterprise/en/wlan/airengine-5761-21-pid-251621259>

2 CloudEngine S5735-L24P4S-A-V2 Switch (PoE)

<https://support.huawei.com/enterprise/en/switch/cloudengine-s5735-l24p4s-a-v2-pid-261313394>

3 CloudEngine S5735-L24T4S-A-V2 Switch (Non-PoE)

<https://support.huawei.com/enterprise/en/switch/cloudengine-s5735-l24t4s-a-v2-pid-261313392>

4 CloudEngine S5735-L48LP4S-A-V2 Switch (PoE)

<https://support.huawei.com/enterprise/en/switch/cloudengine-s5735-l48lp4s-a-v2-pid-261313398>

5 CloudEngine S5735-L48T4S-A-V2 Switch (Non-PoE)

<https://support.huawei.com/enterprise/en/switch/cloudengine-s5735-l48t4s-a-v2-pid-261313396>

6 CloudEngine S5732-H48S6Q Core Switch

<https://support.huawei.com/enterprise/en/switch/cloudengine-s5732-h48s6q-pid-260979645>

7 iMaster NCE-Campus V300R024C00SPC101

<https://support.huawei.com/enterprise/en/nce/imaster-nce-campus-pid-250852420>

13 Notes

A previous version of live heat maps (10.2 from page 74 to 90) have been attached mistakenly. Updated live heat maps will be shared separately.

Proshadi Jayasinghe

Proshadi

14 Amendments

